



Mitigating Algorithm Aversion in Recruiting: A Study on Explainable AI for Conversational Agents

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Abstract

The use of conversational agents (CAs) based on artificial intelligence (AI) is becoming more common in the field of recruiting. Organizations are now adopting AI-based CAs for applicant (pre-)selection, but negative news coverage, especially the black-box character of AI, has hindered adoption. So far, little is known about the contextual factors influencing users' perception of AI-based CAs in general and the effect of provided explanations by explainable AI (XAI) in particular. While research on algorithm aversion provides some initial explanations, information regarding the effects of different XAI approaches on different types of decisions on the attitudes of (potential) applicants is scarce. Therefore, in this study, we use a quantitative, quota-representative study ($n = 490$) to assess the acceptance of CAs in recruiting. By applying an experimental within-subject design, we provide a more nuanced perspective on why and when providing explanations increases user acceptance. We also show that contextual factors such as the type of assessed skills are major determinants of this effect, and we conclude that XAI is not a "one-size-fits-all approach." Based on the insight that contextual factors of the decision problem are more important than the type of XAI approach itself, we argue that the use and the effects of explainability in recruiting need a more nuanced perspective, focusing on the fit of explanations with the user's characteristics and preferences.

Keywords: Explainable Artificial Intelligence; Algorithm Aversion; Recruiting; Experiment; Conversational Agent.

Introduction

Currently, companies face huge challenges in terms of managing their workforce, fueling a war for talent. At the same time, technology trends in e-recruiting reduce switching costs of workers, and as a result, higher volatility and more job posts per company can be observed (Black & van Esch, 2021). In addition, an increasing number of applications for vacancies can be observed (Black & van Esch, 2020). U.K. companies, for example, in 2020 reported an average of 506 applications for low-skilled vacancies and 56 applications for high-skilled vacancies per posted job (CIPD, 2020). However, at the same time, a shortage of qualified candidates among the many applicants per job post is reported, and artificial intelligence (AI) is suggested as a way to identify and activate suitable passive candidates (Black & van Esch, 2021). These trends lead to an expected increase in the use of AI to address these challenges and reduce the workload of human recruiters (Nawaz & Mary, 2019; Sheth, 2018). The increased use of AI is assumed to escalate the

war for talents; thus, successful AI adoption in recruiting will be a competitive advantage (Black & van Esch, 2021). However, actual implementation of AI-based systems in recruiting in the real world remains scarce, with survey data indicating that real-world usage is mainly restricted to prototypes, and its long-term application hardly exists (McKinsey, 2021).

It is well known that opaque information systems (IS), such as black box AI-based systems, have a negative effect on user acceptance (Castelvecchi, 2016). The application of AI in recruiting is considered especially sensitive, as the stakes for applicants are high (Ochmann, Zilker, & Laumer, 2020), and applicants will consequently expect explanations (Kim et al., 2020).

From the legal perspective, recruiting and employment in general were defined as a high-risk use case in the AI act, as proposed by the European Commission (European Commission, 2021). Transparency and giving clear and comprehensible information about the system and its decisions to the user are thus required. Explanations are also seen as required by the European General Data Protection Regulation (Goodman & Flaxman, 2017; Selbst & Powles, 2017) and also by a Californian law (Title 1.81.8. Automated Decision Systems Accountability Act of 2020, 2020) as well as most ethical guidelines on AI (Aslan et al., 2022).

From the technical perspective, approaches to make AI explainable (XAI) and consequently more transparent are proposed (Angerschmid et al., 2022; Dosilovic et al., 2018). The general positive effect of transparency on acceptance has been well established, but how and under what conditions transparency of algorithms influences people's judgment of them remains an open question (Berger et al., 2021).

Literature on algorithm aversion, current regulations, and technology research acknowledge the challenge of deploying AI in recruiting but provide no comprehensive answers about suitable measures to foster adoption. Thus, we start from the technology perspective and investigate the effect of providing explanations in different ways. We investigate (a) how the objectivity of the task, evaluating soft skills as compared to quantifiable qualifications, and (b) the type of XAI explanation, either making the model itself transparent or giving information about relevant decision criteria in the form of post-hoc explanations, influence user acceptance of AI-based systems in recruiting. We thus formulate the following research questions:

1. How do different variants of explanations increase acceptance of AI systems in the field of recruiting?

2. How does the objectivity of the task increase acceptance of AI-based systems in the field of recruiting?

Our results are based on data from an experiment ($n = 490$) with a quota-representative population sample. A scenario design allows us to examine the causal influence of two important factors on acceptance of an AI-based conversational agent (CA) recruiting system: explainability features as an algorithm characteristic and the type of skill that is assessed by the CA as a task characteristic.

Overall, an AI-based recruiting system that delivers a decision with an explanation has been determined to have a higher acceptance than the one that gives a decision without explanation. Among the different AI-based recruiting systems, those that offer explanations based on data referring to verifiable qualifications are more accepted than those that use data on soft skills. We do not find a significant difference in the type of explanation.

The remainder of the paper is structured as follows. In the next section, we develop a research model and our hypotheses. We then describe our measurements and sample. We present the results from these data in the subsequent section. We continue by discussing our results before we present our conclusions in the final section.

A Research Model of Acceptance of AI Hiring Decisions

So far, research has paid little attention to the algorithmic component essential to deal with the mass of information in the context of AI (Kießling et al., 2021). General high-level theories linking system characteristics to acceptance by users, such as the theory of IS success (ISS; DeLone & McLean, 2003), need to be contextualized so that they can be successfully applied to specific phenomena (Hong et al., 2014). Being a more contextualized theory focusing on the algorithmic component, algorithm aversion becomes more and more important in IS research (Burton et al., 2020), in particular, for investigating the adoption of AI in various contexts, such as fake news detection (Kießling et al., 2021), decision support systems (Berger et al., 2021), or recommender systems (Ochmann, Michels, et al., 2020).

Algorithm aversion is discussed as one of the potential causes of the low adoption of AI (Berger et al., 2021; Castelo et al., 2019). A recent literature review by Jussupow et al. (2020) identified several factors influencing user acceptance and incorporated them into a theoretical model of algorithm aversion, which we base our hypotheses on. Characteristics of the

algorithm as the core component of the IS and, if applicable, the human alternative, are proposed as the main influencing factors. For our context of the specific application case of AI in recruiting, we present our research model in Figure 1. However, as research on algorithm aversion in the context of AI remains scarce (Berger et al., 2021) and as currently no measurement instruments for algorithmic aversion are available, we focus on the overall acceptance of AI as one of the measurement approaches that is established in the literature (Jussupow et al., 2020).

One main system characteristic of AI-based systems causing non-adoption has been found in the opaqueness or black-box character of such systems (Castelvecchi, 2016). The literature on algorithm aversion found that a lack of transparency is the major source of human aversion to algorithms in general (Burton et al., 2020; Lehmann et al., 2020). This also fits with the more specific literature on the acceptance of AI in which several researchers found that transparency is an important factor in the acceptance of AI (Dahm & Dregger, 2019; Langer et al., 2019; Mädsche et al., 2019). From the technical perspective, XAI has been introduced as a solution to the black box problem; it was found to increase the acceptance of AI (El Bekri et al., 2020; Toreini et al., 2020). While the fact that transparency increases the acceptance of algorithms and their decisions in various areas has been well established, we also deliver additional support to those findings by replicating them in an experimental study in the sensitive field of recruiting. We, therefore, include explainability as the first algorithm characteristic in our model as having a positive effect on overall acceptance:

Hypothesis 1 (H1): AI-based CA recruiting systems are more accepted when explanations are given as compared to systems without explanations in the context of HR selection and assessment processes.

However, XAI is far away from being a homogeneous technical system, but a general term covering a diverse set of technical approaches. XAI approaches can generally be classified into two categories: first, post-hoc explanations that refer to the output, which give useful information to end-users about why a decision was made (Adadi & Berrada, 2018). Post-hoc explanations explain the outcome of a black box model. They focus on the reasons for the made choice in understandable terms fitting to the mental model of humans, not on the logic of the model itself (Guidotti et al., 2019). Prominent approaches to generate post-hoc explanations of AI models are, for example, Local Interpretable Model-Agnostic Explanations (LIME; Ribeiro et al., 2016) and Quantitative Input Influence (QII; Datta et al., 2016). Results of AI decisions can, for example, be made transparent by presenting a

representative selection of features that are responsible for a decision.

This then enables users to gain insight into a list of features relevant to the outcome of an AI-based system (Ribeiro et al., 2016). This form of XAI was already tested in the context of HR in one published study. Binns et al. (2018) found no effect of different types of explanations on the justice perceptions of an AI decision in the HR context. Over all decision contexts, they find evidence that explanations referring to the functioning of the overall model lead to better justice evaluations, as compared to explanations referring only to the specific case.

Second, there are transparent or intrinsic explanations (in varying levels, see Barredo Arrieta et al., 2020), which allow an understanding of the algorithm itself. Approaches suitable for intrinsic explanations include linear models, decision trees, and rules (Rudin, 2019; Samek, 2020). Explanations based on decision trees are particularly important due to the simplicity of their interpretation. They are thus a popular model used in the context of decision support systems as they allow an understanding of how the model uses features in making predictions (Letham et al., 2015; Rudin, 2019). Such transparent models have been argued to be superior to post-hoc explanations of black box models in terms of understanding the decision (Rudin, 2019).

Summing up, studies investigating the impact of XAI on acceptance so far found mixed results. Furthermore, only one of the two main types of XAI approaches has been given attention so far. This leaves the important question of how those two main types of XAI are related to the acceptance of AI-based IS. Literature on algorithm aversion is one major research stream in IS to explain limited acceptance in IS that we can base a hypothesis on in this regard. Scholars argue that an increased understanding of the underlying workings of a system can lead to higher levels of trust and thus reduce algorithm aversion (Yeomans et al., 2019). On the one hand, an understanding of the inner workings can be formed from repeated interactions with the system and may even lead to blind trust or automation bias (Jacovi et al., 2021). On the other hand, explanations making the underlying mechanisms of the system transparent can also provide such an understanding, especially when no previous interactions with the system took place (Hoff & Bashir, 2015). Intrinsic explanations aim at providing this overall understanding of the underlying mechanisms, rather than just justifying one single decision. In this case, a better understanding of the inner workings of a system helps foster acceptance and use (Seong & Bisantz, 2008). J. D. Lee and See (2004) even argue that trust in IS is only deemed necessary when a complete understanding is either impractical or not possible at all.

This theoretical perspective on trust generation allows us to formulate a concrete hypothesis on the effect of transparent models as compared to post-hoc explanations of black box models. In the case of intrinsic explanations, a clear understanding of the underlying mechanisms is deemed possible and should thus result in the greatest acceptance of the system. In the case of provided post-hoc explanations, the understanding of a single decision but not of the underlying mechanisms and rules can be achieved. We, therefore, assume an effect of the type of XAI on overall acceptance in our contextual model and formulate our second hypothesis as follows:

Hypothesis 2 (H2): AI-based CA recruiting systems offering intrinsic explanations have a higher acceptance than systems offering post-hoc explanations in HR selection and assessment contexts.

The concrete implementation of explainability in AI-based systems cannot be assessed independently from the characteristics of the decision task. When those specific task characteristics match the technology used to accomplish it, this results in a high task-technology fit, i.e., the technology is (perceived to be) able to support performing the task (Goodhue & Thompson, 1995; McGill & Klobas, 2009). Concretely, an important characteristic of the task is the level of objectivity related to the assessment of decision factors in general. It has been shown that for objective tasks in the form of quantitative analysis, algorithms are perceived as more suitable, as compared to subjective tasks requiring intuition and interpretation (Castelo et al., 2019). This has also been examined through the lens of the type of skill, either being “emotional” or “cognitive” judgments,¹ wherein a behavioral tendency to ascribe less emotional capacity to robots as compared to humans has been shown (Waytz & Norton, 2014). M. K. Lee (2018), who studied the evaluation of AI decisions in comparison to decisions made by humans, partly also in the context of HR, found that for “mechanical” tasks, AI decisions were rated similar to decisions made by humans. For tasks requiring human skills, however, AI decisions were evaluated worse than human decisions.

For the case of decisions in the recruiting process, specifically, it has been demanded that explanations should refer to a match between the criteria of the job’s requirement profile and the skills of the applicant (Schuler, 2013). Requirement profiles of a job can be structured along the knowledge, skills and abilities, behavioral competencies, qualifications and training,

experience, specific demands, and other special requirements needed to perform the tasks required by the position (Armstrong & Taylor, 2020). From the assessment perspective, human or mechanical skills can be distinguished. For example, qualifications and training refer to “professional, technical, or academic qualifications required or the training that the candidate should have undertaken” (Armstrong & Taylor, 2014), and its assessment requires mostly mechanical skills. Behavioral competencies, on the other hand, refer to soft factors that humans need to interpret in order to assess the fit to a team or an organization’s culture.

For this reason, task characteristics come into focus when considering XAI, and a perceived task-technology fit is expected to increase the overall acceptance of the IS. Especially, the objectivity of tasks has been shown to affect acceptance. We hypothesize that the use of more objective decision criteria by an AI-based IS results in a better task-technology fit and thus has a higher overall acceptance.

Hypothesis 3 (H3): AI-based CA recruiting systems offering explanations referring to verifiable qualifications have a higher acceptance than systems offering explanations referring to soft skills in HR selection and assessment contexts.

Method

To investigate the influence of XAI approaches on the acceptance of AI in preselection decisions, we use a vignette-style method, which is suited to be combined with an experimental design in questionnaire surveys (Araujo et al., 2020). Experimental vignette-style studies have been used previously to study AI decisions (M. K. Lee, 2018). Following this approach, we designed an experiment and presented subjects with scenarios that are varied by (a) the type of XAI approach and (b) the level of objectivity, that is, referring to measurable qualifications or soft skills, allowing us to test our hypotheses. For each vignette, subjects evaluate their overall acceptance of such an AI-based system. We then evaluate the AI-based recruiting systems with different CA decision features. Thus, we evaluate the influence of this system feature on the overall system acceptance. Note that we do not provide our subjects with a complete mimicked application process including a chat with a CA, but rather provide them with static vignette pictures mimicking the outcome of such a process.²

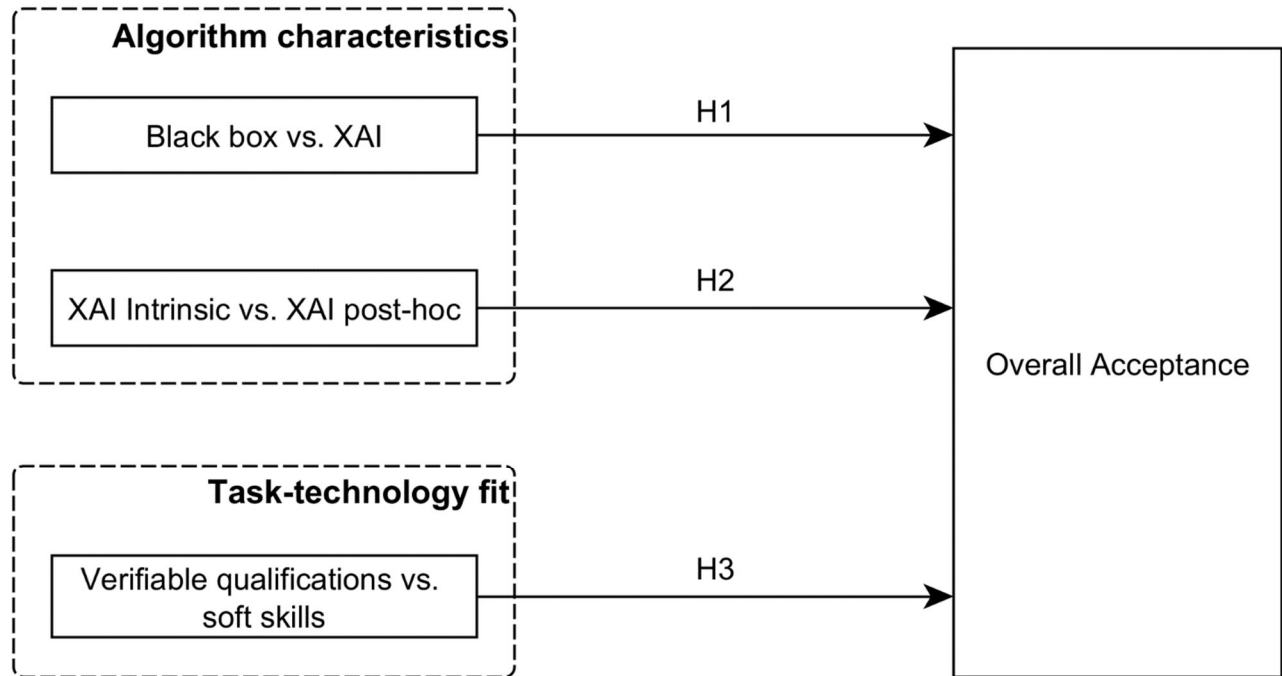


Figure 1. Research Model

Study and Scenario Design

Before participants were presented with the scenarios, they were given a general introduction to the context. Specifically, the participants should put themselves in a situation in which they want to apply for a job and click “apply now” on the job advertisement; a CA instead of a human recruiter then performs the preselection. The CA would communicate with them by chat and ask all the necessary questions to assess their fit for the open position.

After this introduction, seven scenarios were subsequently presented to the participants in random order. In all scenarios, participants were informed that they were rejected by the CA in the preselection process. In our baseline scenario, BASE, participants were simply informed that the CA decided to reject their application. This mimics the result-focused decision of a typical AI-based CA recruiting system based on a (black box) AI without an XAI component.

We vary BASE according to two factors: explainability (EXPLAIN) and type of skill representing the objectivity that the decision is based on (SKILLTYPE, for testing Hypothesis H3). Following the distinction between post-hoc explanations of black box models and intrinsic explanation approaches laid out in Section 2, we select three variants for the factor EXPLAIN. EXPLAIN_LIST is an example of a post-hoc explanation, as it names the relevant criteria but does not make the decision logic of the model itself

transparent. It mimics feature-focused techniques, such as LIME, by providing participants with a list of three criteria that were relevant for the rejection. The second variant, that is, EXPLAIN_COMPARE, is also a post-hoc explanation and provides a visualization of the score that the CA assigned to them in the three features relevant to the decision of EXPLAIN_LIST and also provides the average score of other applicants. In EXPLAIN_INTERPRET, subjects are shown a simple decision tree using the same criteria as in the first two variants of EXPLAIN and also making the decision logic of the model transparent. The chosen decision tree is simple enough that a full understanding of the underlying ruleset can be assumed to be possible (Ramon et al., 2021; Samek, 2020). The path to the decision “reject” is also highlighted to facilitate the replication of the decision without the need for a mathematical background (Barredo Arrieta et al., 2020).

We again vary all three EXPLAIN variations with two variations of the factor SKILLTYPE, resulting in a three-by-two design. In our field of application of recruiting, we incorporate the former as soft skills in SKILLTYPE_SOFT and the latter as more objectively verifiable qualifications in SKILLTYPE_VERIFIABLE. For soft skills, we use the ability to work in teams, communication skills, and diligence, whereas for verifiable qualifications, we use work experience, command of English, and computer knowledge. The VERIFIABLE criteria are exemplifications of the

qualifications and training category of the KSA model, while the SOFT criteria fall under behavioral competencies (Armstrong & Taylor, 2020).

Combining each of the three EXPLAIN variations with each of the two SKILLTYPE variations results in six combinations; we, therefore, constructed six treatment scenarios in addition to BASE. We refer to each of these six treatment scenarios by combining the variation names of the EXPLAIN and SKILLTYPE factors, respectively. For example, in the interpretable model in the form of a decision tree, INTERPRET,

using soft skills as qualifications, SOFT, is referred to as INTERPRET_SOFT scenario. All treatment scenarios and the corresponding key elements of the explanations as they were shown to the participants are displayed in Table 1. The comparison of all EXPLAIN scenarios with BASE allows for the testing of Hypothesis H1, while a separate analysis of the EXPLAIN_INTERPRET scenarios with the other EXPLAIN scenarios can test H2. Finally, a comparison between the SOFT and VERIFIABLE scenarios serves as the test for Hypothesis H3.

Table 1. Overview of Scenarios

		EXPLAIN																		
		LIST	COMPARE	INTERPRET																
SKILLTYPE	SOFT	Ability to work in teams Communication skills Diligence	Soft skills <table><thead><tr><th></th><th>bad</th><th>average</th><th>good</th></tr></thead><tbody><tr><td>Ability to work in teams</td><td>● You</td><td>● Average of other Applicants</td><td></td></tr><tr><td>Communication Skills</td><td></td><td>● You</td><td>● Average of other Applicants</td></tr><tr><td>Diligence</td><td></td><td>● You</td><td>● Average of other Applicants</td></tr></tbody></table> ● You ● Average of other Applicants		bad	average	good	Ability to work in teams	● You	● Average of other Applicants		Communication Skills		● You	● Average of other Applicants	Diligence		● You	● Average of other Applicants	Your Result: Ability to work in teams bad average or good Diligence bad average or good Communication Skills average or bad NO YES
		bad	average	good																
Ability to work in teams	● You	● Average of other Applicants																		
Communication Skills		● You	● Average of other Applicants																	
Diligence		● You	● Average of other Applicants																	
VERIFIABLE	Work Experience Command of English Computer Knowledge	Verifiable Qualifications <table><tbody><tr><td>Work Experience</td><td>0 Years</td><td>● You</td><td>● Average of other applicants</td><td>20 Years</td></tr><tr><td>Command of English</td><td>None</td><td>● You</td><td>● Average of other applicants</td><td>Expert</td></tr><tr><td>Computer Knowledge</td><td>None</td><td>● You</td><td>● Average of other applicants</td><td>Expert</td></tr></tbody></table> ● You ● Average of other applicants	Work Experience	0 Years	● You	● Average of other applicants	20 Years	Command of English	None	● You	● Average of other applicants	Expert	Computer Knowledge	None	● You	● Average of other applicants	Expert	Your result: Work Experience less than 5 years 5 years or more Command of English lower than C1 Equal or higher than C1 Computer Knowledge below average average or higher NO YES		
Work Experience	0 Years	● You	● Average of other applicants	20 Years																
Command of English	None	● You	● Average of other applicants	Expert																
Computer Knowledge	None	● You	● Average of other applicants	Expert																

Note: The table displays our own translation of the original explanations in German that were given to participants for the CA's preselection rejection decision. The total of six explanations (i.e., treatment scenarios) results from the combination of the three EXPLAIN and the two SKILLTYPE variations. The complete translated questionnaire can be found in the Appendix.

Measures

As a dependent measure, we use a scale termed overall acceptance, covering multiple aspects of technology acceptance (Gaudiello et al., 2016; Ghazizadeh et al., 2012). Concretely, we measure *intention to use* (three questions adapted from Venkatesh et al., 2003), *trust*, and *fairness perception* (one question each in line with earlier research, e.g., Araujo et al., 2020; M. K. Lee, 2018). Concretely, the following questions were asked, with the shorthand notations used in the results section added at the end in parentheses:

- The use of a chatbot during the application process [that explains its decision using varying according to the scenario, none in BASE] positively influences my decision to apply. (Applyplus)
- If this chatbot is available as an alternative to a human interviewer, then I choose the chatbot. (Choosebot)
- I consider the use of chatbots in the application process to be positive. (Goodbot)
- Do you trust the chatbot decision just described? (Trust)
- How fair or unfair is this preselection decision of the chatbot? (Fair)

For all the scenarios, the resulting acceptance scales show very high reliability with Cronbach's alpha (α) values between 0.94 and 0.96. Eigenvalues from exploratory factor analyses for all scenarios are above 4.0 for the first factor and below 0.5 for the second factor, strongly indicating a one-factor solution and thus the computation of a single scale for those five items. In addition, we have also measured if our subjects consider it important for each scenario that a human recruiter checks this decision of the chatbot. Table 2 shows descriptive statistics for all dependent variables and overall acceptance for each scenario separately. For all scales, higher values indicate more agreement rated on a seven-point scale.

As independent control variables, we measure several sociodemographic variables and two scales. The items of the two scales were measured using a five-point scale from one to five, wherein higher values indicate stronger agreement. *Affinity for technology interaction* (ATI), consisting of nine items developed by Franke et al. (2019), measures the tendency to actively engage with technology ($\alpha = 0.86$, $M = 3.05$, $SD = 0.84$). *Trust in AI* was measured with the scale developed by Jian et al. (2000) and translated into

German and modified by Pöhler et al. (2016), where "system" was replaced by "AI system" ($\alpha = 0.87$, $M = 2.91$, $SD = 0.61$). This construct was used due to the widespread acceptance of the English version and its applicability in several domains where automated activities take place. Table A1 in the Appendix provides the descriptive statistics for all individual questions of those scales.

Subjects and Procedure

The vignette-style study was implemented as a lab-in-the-field study (Gneezy & Imas, 2017) using an online questionnaire in Unipark.³ Subjects were recruited from an ISO 26362 certified online access panel of the survey company Norstat between January 13 and February 12, 2020.⁴ In total, 558 subjects were recruited as a quota-representative population sample regarding age, education, and gender for Germany and Austria ($n = 220$ for Austria and $n = 338$ for Germany). Two subjects completed the questionnaire twice and were thus removed from the dataset, leaving us with 554 observations. We also excluded eight subjects who completed the survey after having stopped and restarted it at a later point in time. Looking at the completion times of those 546 subjects who completed the survey in one pass, we find and remove outliers who completed the survey very fast (mean duration of 235.4 seconds for the fastest 5 percent) or very slow (mean duration of 2,903.7 seconds for the slowest 5 percent). This leaves us with a total of 490 subjects, who, on average, took 748.9 seconds to complete the survey ($SD = 343.8$ seconds). All results presented in Section 4 are based on those 490 subjects (190 from Austria and 300 from Germany). For a robustness check of those results with regard to the exclusion and inclusion of the subjects with the top and bottom 5 percent completion times yielding comparable results, see the Appendix.

The survey consisted of two parts. In the first part, subjects completed the survey scales for general trust in AI and affinity for technology, as well as additional questions. In the second part, subjects were introduced to the context of the scenarios. They then were shown and evaluated each of the scenarios described in Section 3.1, in random order. We, therefore, use a within-subject design where each subject evaluated each of the scenarios, to also be able to detect smaller effect sizes. Figure 2 depicts the experimental sequence as well as the results of the randomization for the first scenario. All subjects were informed that they were presented with predefined scenarios that are purely hypothetical.

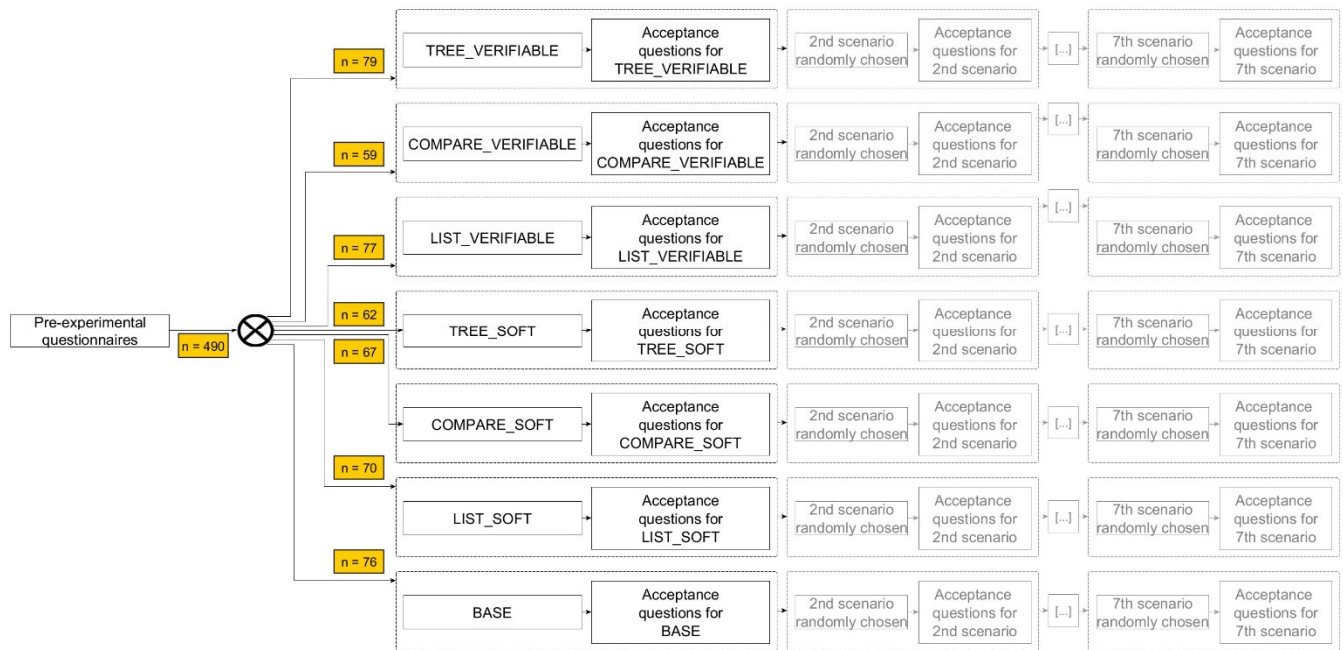


Figure 2. Structure of the Experiment

Note: The figure displays the structure and timeline of the experiment, including the first steps in the paths taken in the randomized presentation of the scenarios.

Table 2. Descriptive Statistics for Dependent Variables by Scenario

Dependent Variables	BASE (No Info)	EXPLAIN					
		LIST		COMPARE		INTERPRET	
		VERIFIABLE	SOFT	VERIFIABLE	SOFT	VERIFIABLE	SOFT
Applyplus	3.44 (1.70)	3.93 (1.69)	3.77 (1.70)	4.03 (1.71)	3.81 (1.75)	3.99 (1.65)	3.77 (1.71)
Choosebot	3.39 (1.75)	3.61 (1.77)	3.50 (1.79)	3.73 (1.79)	3.53 (1.75)	3.66 (1.72)	3.52 (1.76)
Goodbot	3.55 (1.68)	3.81 (1.70)	3.69 (1.72)	4.00 (1.75)	3.77 (1.71)	3.88 (1.71)	3.73 (1.70)
Trust	3.43 (1.55)	3.82 (1.60)	3.56 (1.62)	3.91 (1.61)	3.67 (1.64)	3.87 (1.62)	3.62 (1.62)
Fair	3.63 (1.47)	3.96 (1.53)	3.74 (1.48)	4.07 (1.52)	3.83 (1.55)	3.97 (1.51)	3.78 (1.51)
Overall	3.49 (1.48)	3.83 (1.53)	3.65 (1.53)	3.95 (1.55)	3.72 (1.57)	3.87 (1.51)	3.68 (1.54)
Alpha	0.94	0.95	0.95	0.96	0.96	0.95	0.96

Note: The table displays the mean values (standard deviations in parentheses) for all dependent variables in each scenario. All items range from one to seven, with higher values indicating more trust, agreement, etc. “Overall” refers to the overall acceptance scale and is computed as the mean value of the behavioral intention, trust, and fairness measures. The bottom row labeled “Alpha” contains the Cronbach’s alpha values for the overall acceptance scale in each of the scenarios separately, with all values indicating very good consistency. All values are based on n = 490 participants. Inclusion of participants at the upper and lower tails of the duration distribution results in only minor changes to the means and standard deviations (see Table A4 in the Appendix).

Results

We begin our analysis by investigating the mean values in our dependent variables for all scenarios, as shown in Table 2. As per the results, we were able to determine the lowest mean values for all dependent variables for the baseline scenario without explanation, BASE. Here, participants were found to least likely state that the usage of a CA without explanations increases the likelihood to apply for a job (M = 3.44),

to choose this CA over a recruiter (M = 3.39), to consider the usage of such a CA to be a good thing (M = 3.55), to trust in the decision of this CA (M = 3.43), to consider this a fair decision (M = 3.63). Accordingly, the overall acceptance scale is similarly low, with a mean value of M = 3.49 on a scale ranging from one to seven.

To test the effect of the different forms of explainability in our scenarios on the overall acceptance scale, we

estimate a mixed linear model. We include subject random effects, as the 3,430 decisions made in the scenarios are nested within 490 subjects (i.e., each respondent completed all seven scenarios in random order). The model is estimated using the maximum likelihood estimator (lmer function from the R package lme4 (Bates et al., 2015) and lmerTest for the calculation of the p-values (Kuznetsova et al., 2017)). The use of the mixed linear model avoids the underestimation of the standard errors (and the corresponding p-values) of the coefficients that would occur when using simple OLS models for nested data.

Table 3. Multilevel Regression Models on Overall Acceptance of CA Preselection Decision

EXPLAIN Vignette		Model 1
VERIFIABLE	LIST	0.338 (8.431)***
	COMPARE	0.458 (11.439)***
	INTERPRET	0.382 (9.532)***
SOFT	LIST	0.161 (4.027)***
	COMPARE	0.231 (5.781)***
	INTERPRET	0.195 (4.873)***
Controls		
Affinity for Technology		-0.021 (-0.282)
Trust in AI		0.669 (12.077)***
Gender (1 = Female)		-0.005 (-0.040)
Country (1 = Austria)		-0.019 (-0.165)
Age		-0.007 (-2.089)*
Income		-0.007 (-0.307)
(Intercept)		-0.547 (-1.284)
AIC		8146.189
Δ AIC		3.328
Log Likelihood		-4058.059
Number of Decisions		3430
Number of Participants		490

Notes: ***p < 0.001; **p < 0.01; *p < 0.05. The table displays the unstandardized coefficients of a multilevel regression model with subject random effects estimated using the maximum likelihood estimator and the overall acceptance scale as the dependent variable. t-values are displayed in parentheses. Dummy variables are included for all six EXPLAIN scenarios with BASE serving as reference category; the dummy variables take the value of one for subjects' answers in the vignette referenced in the variable name and zero for all other observations. The regression coefficient of a scenario dummy can be interpreted as the average difference in overall acceptance of this vignette to the reference vignette base.

Model 1 in Table 3 includes six scenario dummies as fixed effects with BASE as the reference category, as well as the control variables introduced in the previous section.⁵ We were able to determine significant effects for all six EXPLAIN scenario dummies, indicating, together with the positive regression coefficients, that

in all EXPLAIN scenarios, subjects score higher on the overall acceptance scale compared to subjects' decisions in BASE (all p < 0.001).

This introductory result replicates the current state of the literature on the general positive effect of explanations, as we find a positive effect for each scenario introducing explanations compared to base without explanation. This constitutes evidence in favor of Hypothesis 1: Explanations are found to increase acceptance of an AI-based CA recruiting system compared to a system offering no explanations.

To further investigate the differences between the scenarios, we look at the Tukey-corrected p-values from pairwise comparisons between the scenarios (R package "emmeans" by Lenth et al. (2020)). The comparisons again show that all six EXPLAIN conditions are significantly different from the BASE condition (all p-values $\leq$ 0.001, the p-values of all pairwise comparisons are displayed in Table 4). More interestingly, we find that all three SOFT conditions do not differ significantly from each other. Each of the three EXPLAIN_SOFT scenarios was found to differ significantly from each EXPLAIN_VERIFIABLE scenario, with one exception: EXPLAIN_COMPARE_SOFT does not differ from EXPLAIN_LIST_VERIFIABLE (p = 0.112). As for the comparisons among the three EXPLAIN_VERIFIABLE conditions, only the difference between EXPLAIN_COMPARE_VERIFIABLE and EXPLAIN_LIST_VERIFIABLE is significant (p = 0.043).

To compare the magnitude of the overall acceptance between the EXPLAIN scenarios, we compute 95-percent confidence intervals for the regression coefficients of the EXPLAIN dummies from Model 1 using 1,000 parametric bootstrap samples (function confint.merMod in the lme4 R package (Bates et al., 2015)), as displayed in Figure 3. As all coefficients of the scenario dummies are significantly different from zero, the lower bounds of the confidence intervals accordingly are larger than zero. We observe the strongest effect, compared to BASE, for COMPARE_VERIFIABLE with a point prediction of 0.46 and the confidence interval ranging from 0.38 to 0.53. This explanation of a CA's decision was able to increase the overall system acceptance with up to 0.53 points on a seven-point scale. With point predictions of 0.38 and 0.34, TREE_VERIFIABLE and LIST_VERIFIABLE, together with COMPARE_VERIFIABLE, form a cluster of, compared to the EXPLAIN_SOFT conditions, stronger effects. The three EXPLAIN_SOFT conditions form a cluster of comparatively weaker, albeit significantly different from zero, effects. This is consistent with our Hypothesis 3, which stated that AI-based CA recruiting systems offering explanations referring to verifiable

qualifications have a higher acceptance compared to systems offering explanations referring to soft skills. At the same time, Hypothesis 2, stating that intrinsic explanations have a stronger effect than post-hoc explanations, is thus not supported.

The tentative result that the VERIFIABLE scenarios referring to hard facts of qualifications show a stronger effect than the scenarios that base their explanations on more vague soft skills illustrates that XAI is not a simple one-size-fits-all approach but rather depends on different contextual factors. In this case, referring to

more “mechanical” tasks as opposed to “soft skill” tasks increases the effect of explanations regardless of the type of XAI used. As for the type of XAI used, transparent models versus post-hoc explanations, both theories and assumptions made in previous publications indicate that the former should be seen as superior since a complete understanding of the model should have a stronger effect than a post-hoc explanation only giving hints on the inner workings of the model. Surprisingly, no difference was noted between the two variants.

Table 4. Pairwise Comparisons

			BASE No Info	EXPLAIN					
				LIST		COMPARE		INTERPRET	
				VERIFIABLE	SOFT	VERIFIABLE	SOFT	VERIFIABLE	SOFT
EXPLAIN	LIST	VERIFIABLE	< 0.001						
		SOFT	< 0.001	< 0.001					
	COMPARE	VERIFIABLE	< 0.001	0.043	< 0.001				
		SOFT	0.001	0.112	0.581	< 0.001			
	INTERPRET	VERIFIABLE	< 0.001	0.928	< 0.001	0.477	0.003		
		SOFT	< 0.001	0.007	0.980	< 0.001	0.972	< 0.001	

Note: The table presents the Tukey-adjusted p-values of pairwise comparisons between all conditions computed using the R package emmeans (Lenth et al., 2019).

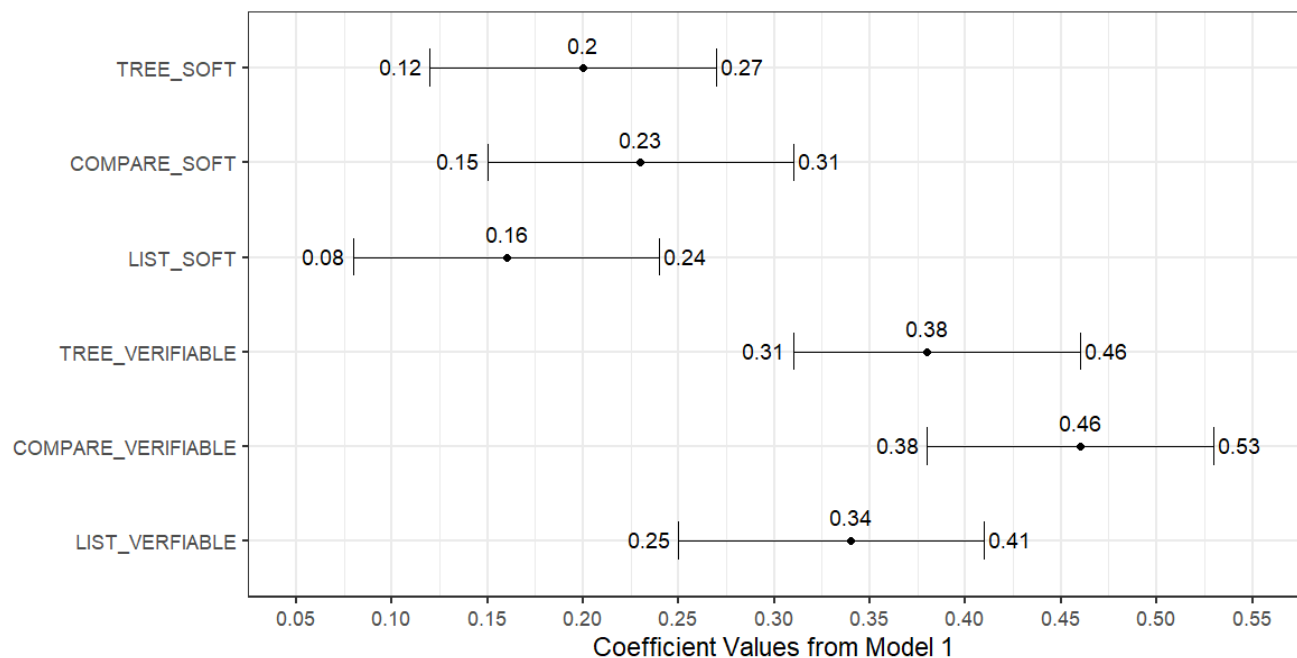


Figure 3. Coefficients and Confidence Intervals for Scenario Dummies

Note: The figure displays the regression coefficients of the scenario dummies from Model 1 with the corresponding 95-percent confidence intervals, computed using 1,000 parametric bootstrap samples.

For the control variables ATI and general trust, a positive effect with a coefficient of 0.669 ($p < 0.001$) was noted for the general trust in AI and no significant effect for affinity for technology. For the sociodemographic control variables gender, age, income, and a country dummy for Germany in comparison to Austria, we only find a significant negative effect of -0.007 ($p < 0.05$) for age.

Discussion

In this section, we briefly discuss our results in light of the literature, the implications for extant research, and finally the implications for practitioners.

Discussion of Results

The results on Hypothesis 1 show the general positive effect of explanations. The rejection of Hypothesis 2 shows that, in contrast, the type of explanation is of less relevance in first encounter situations, surprisingly contradicting both previous views expressed in the literature and assumptions of algorithm aversion theory. In this regard, it seems questionable to build the required understanding of the inner working of the algorithms just by providing more transparent models, that is, decision trees. However, the reduction of algorithm aversion through explanations is highly dependent on contextual factors, such as assessed skills of the underlying decision problem. The results regarding Hypothesis 3 show that explanations based on verifiable facts lead to higher acceptance than explanations on soft factors.

We summarize these main insights and their relation to our hypotheses and the state of the art in the literature in Table 5 before we discuss their implications for theory and practice in the two subsequent sections.

Implications for Theory

In this study, we showed that the positive effect of explanations based on both transparent models and post-hoc explanations on acceptance is not independent of contextual factors. This contradicts arguments from the literature that transparent models are generally preferred as they allow for a full understanding of the model, while post-hoc explanations can be understood as only approximations of the original model (Rudin, 2019). Second, we demonstrated that the contextual factor “data type used for decision-making” is more important than the type of XAI used. So far, several researchers highlighted inconclusive results in this regard. We contributed to this research stream as we found first evidence that contextual factors are more important

than technical features, as in the case of recruiting. While we find no significant differences between the XAI approaches, a post-hoc power analysis suggests there is a small effect that could be detected with larger sample sizes (power analysis available in the Appendix).

From the perspective of HR research, our results replicated the finding that for rejection decisions made by humans, explanations were also found to elicit a more positive reaction than simple information about a rejection (Bies & Shapiro, 1988). Decisions made by AI-based CA recruiting systems typically replace decisions that were traditionally made by humans, and those decisions have often already been examined extensively in HR research. It is evident that research on XAI in recruiting, but also in general, can learn from studies on the AI's human counterpart. In this regard, we delivered first evidence that the properties of decision factors are also relevant for AI-based CA, as they may influence the effect that an explanation has on the user.

One key insight in this regard is that the level of objectivity for the skill assessment significantly influences user acceptance, that is, explanations referring to measurable qualifications increase acceptance more than explanations referring to soft factors. This finding again follows a pattern of results established in a meta-analysis on applicants' reactions in traditional hiring processes, where standardized cognitive tests are found to be better accepted than personality tests (Truxillo et al., 2009). Therefore, explanations themselves are a necessary but not a sufficient condition to increase acceptance. Task-technology fit is also deemed necessary (McGill & Klobas, 2009). Explanations have to be meaningful to the user and accepted by the user for the decision at hand and its characteristics. Our study provides first evidence for soft skills and verifiable qualifications. This is an important insight from a design perspective and thus calls for more research on how to embed XAI in real-world processes. In our case, the insight could be used to analyze the recruiting process regarding the objectivity of skills assessed in various phases of the recruiting process. Finally, CAs could be primarily deployed for recruiting subprocesses focusing on data capturing measurable qualifications. Those insights on contextual factors in embedding XAI in real-world processes are needed to specify the application of CAs in practice more clearly. IS research showed that those contextual and integration factors of IS deployment often are more important than technical features (see, e.g., Cubric, 2020; Felzmann et al., 2019; Floridi et al., 2020). Thus, we delivered first evidence that the type of data a recruiting decision is based on is an important factor to deploy CAs and XAI.

Table 5. Hypotheses and Contributions

Hypotheses	Answers in the literature	Our contribution
AI-based CA recruiting systems are more accepted when explanations are given as compared to systems without explanations in the context of HR selection and assessment processes.	Algorithm aversion literature established that the black-box character of AI-based IS hinders trust and acceptance of such systems and that explanations in general can increase acceptance (Langer et al., 2019; Mädche et al., 2019).	We provide a more nuanced perspective on this effect, showing that contextual factors such as the assessed skills are major determinants of this effect.
AI-based CA recruiting systems offering intrinsic explanations have a higher acceptance than systems offering post-hoc explanations in HR selection and assessment contexts.	Transparent models are assumed to provide a superior understanding of the inner workings of AI-based IS (Rudin, 2019). Algorithm aversion assumes that a better understanding of the inner workings of an AI-based IS increases acceptance when trust cannot be built over time (Hoff & Bashir, 2015).	We show that this hypothesized effect of explanations based on simple transparent models compared to post-hoc explanations cannot be found. We show that intrinsic and transparent models are not per se superior compared to post-hoc explanations. It seems challenging to get a sufficient understanding of the initial workings in an initial encounter with the system.
AI-based CA recruiting systems offering explanations referring to verifiable qualifications have a higher acceptance than systems offering explanations referring to soft skills in HR selection and assessment contexts.	A high task-technology fit has been shown to increase the acceptance of technologies (Goodhue & Thompson, 1995; McGill & Klobas, 2009). Precursor studies showed that AI is more accepted for mechanical tasks as compared to tasks seen as requiring human cognition (Castelo et al., 2019; M. K. Lee, 2018; Waytz & Norton, 2014). The HR literature has shown that standardized tests are more accepted for hard facts as compared to soft skills (Bies & Shapiro, 1988; Truxillo et al., 2009).	We show that task characteristics in the form of assessing verifiable qualifications have a higher acceptance as compared to assessing soft skills. This provides an empirical test of the task-technology fit hypothesis for AI-based IS. Further, we show that the use and the effects of explainability in recruiting need a more nuanced perspective focusing on the fit of explanations with the user's mental model and preferences.

In a related manner, self-determination theory has previously been used to assess motivation to use an IS, finding that autonomy-supportive IS, as compared to controlling IS, increases the motivation to use it by creating a positive and also meaningful experience (Peters et al., 2020; Yang & Aurisicchio, 2021). For an autonomy-supportive IS, the psychological needs of competence, relatedness, and autonomy need to be met (Vreede et al., 2021). Especially competence, as the need to have the ability to achieve the desired outcome when interacting with an IS, and autonomy, as the need to be able to make my own choices rather than having them made for me by the IS, are of relevance in the context of our study. This means the explanations the CA gives fit the user's competence and that the user has the autonomy to explore the feedback in various ways. The second implication of

self-determination theory leads to additional complexity by conceptualizing the suitability of explainability as an individual difference. This has recently been shown for explanations of AI decisions and different cognitive styles (Ramon et al., 2021). Taken together, those implications following from our results in conjunction with self-determination theory further contribute to our argument that explanations of algorithmic decisions must account for the context in which they are made and ideally are also tailored to different user groups.

Managerial Implications

This study provides several practical implications. First, XAI can increase the acceptance of AI-based CA recruiting systems and has implications for

organizations adopting CAs and AI in recruiting. While previous research does imply a positive effect of using AI in recruiting (Hunkenschroer & Luetge, 2022), our results show that XAI can help mitigate potential negative effects arising from black box AI without explanations. Organizations should also follow general recommendations from HR research that found that providing explanations to rejected applicants may influence not only organizational attractiveness but also their reactions to the decisions. Applicants were found to be more likely to recommend the organization to others when they received an explanation about the selection decision (Gilliland, 1994; Truxillo et al., 2009). Consequently, despite the development of AI-based CAs in general and as a tool in the recruiting process, organizations need to address issues such as fair, accountable, and transparent (FAT) AI to improve applicants' overall acceptance as it can foster trust, intention to use, and finally the number of applicants.

Second, the current study shows that the effect of the type of explanations on acceptance of AI-based CA recruiting systems depends on the decision variables. Explanations referring to measurable skills increased acceptance more than explanations based on soft skills. It thus seems advisable to conduct an in-depth analysis of the recruiting process before adopting CAs. Especially, current practice in candidate preselection, that is, creating a shortlist of candidates meeting the formal criteria, seems a promising first step for applications where AI can offer benefits over traditional systems (Simperl & Mochol, 2009). Also, a stepwise implementation starting with known elements, like quantifiable hard skills for preselection, might pave the way for future broader applications.

Third, our study showed the high importance of contextual factors when providing explanations in recruiting. The practical implication of this work concerning the design of CAs for recruiting is that organizations should aim for developing IS that are tailored to their application context and provide meaningful and relevant explanations, as there remains a gap in evaluating user-centric explanations (Förster et al., 2020). To address the question of which explanation feature(s) potential applicants require and how they should be designed, the IS community is challenged to make relevant contributions toward designing CAs based on XAI in recruiting.

Limitations and Future Research

Our results are not without limitations that must be considered when thinking about their implications. However, these limitations and our study, in general, can serve as a starting point for future research on AI-based CA recruiting systems.

One limitation of this research is that we did not offer in-depth explanations for being rejected to the study participants. We selected three common technical-driven approaches of explanation discussed in the literature. Nevertheless, there exist several other XAI approaches and ways to present explanations. However, in our view, existing XAI approaches have mostly been researched and discussed from a technical perspective and treated in isolation, commonly without considering recommendations of approaches taking context factors and, even more important, specific contexts such as recruiting into account.

Förster et al. (2020) found that humans appreciate concreteness, coherence, and relevance as characteristics of automatically generated explanations. They also found that longer explanations are appreciated more by users than shorter explanations. This points to another limitation of our study; while we differentiated between three different XAI methods, as the literature suggests that the presentation of results may significantly influence user perception (Huysmans et al., 2011), we have not examined varying degrees of detail in our explanations, which is a promising avenue for future research.

Subjects in our study were presented with hypothetical scenarios, which do not involve them as directly as would be the case in a real-world, high-stakes scenario. Nevertheless, our study provides important findings that can serve as a starting point for future research. Similarly, the explanations of the rejection decision were not created from a real AI-based CA recruiting system but were designed as the presentation of the components in each scenario that were visualized in a form understandable for laypeople, which we considered important. Similarly, the questionnaire that subjects completed before they rated the scenarios might have influenced their ratings. However, a currently unpublished follow-up study with questionnaires presented *after* the scenarios replicates the mean difference for the BASE and a variant of the COMPARE_VERIFIABLE scenario.⁶ Regarding the issue of hypothetical scenarios having a weaker effect compared to real-world situations, the issue of deception arises with the use of our research design. While we did inform all participants that the scenarios were purely hypothetical, they were not informed beforehand that all participants would receive a rejection decision, that is, we hold the outcome constant over all subjects and scenarios. This can be construed as deception. We, however, opted not to inform the subjects about this fact as this knowledge would have an unpredictable impact on the results, which would make a clear interpretation difficult.

In a related manner, to best test our hypotheses, we used a black box AI as our reference scenario, while in most real-world cases, the natural alternative to using an AI is the use of a human recruiter. Similarly, we included the comparison with the assessment of others in the listed skills only and not for the transparent models. The inclusion of such a comparison in the INTERPRET scenarios could potentially lead to an increase in their acceptance. Moreover, we only examined rejection decisions backed by three criteria. In real-world recruiting situations, various criteria can determine hiring decisions, which were not studied here, as well as how the results would change if subjects interacted with a real CA before the evaluation. Finally, only one type of AI-based CA recruiting system and its effect on job seekers was studied. Those limitations pave the way for a host of potential studies to examine the stability of our results in various conditions.

In addition, when comparing Model 1, a mixed linear model based on all 3,430 decisions made in the within-subject experiment, with the OLS Model R1 in the Appendix, based on only the first decision that each subject made (which corresponds to a between-subject approach as scenarios were presented to each subject in random order), we find that while the coefficients are smaller but qualitatively similar, they are insignificant. Concretely, simulations indicate that a between-subject approach would need a larger sample size to result in sufficient statistical power to reliably detect the effects we have found. We gauged the strength of our effects using our regression coefficients, which can be seen as raw effect-size measures (Funder & Ozer, 2019). As those are point estimates, we complemented them with bootstrapped confidence intervals, as illustrated in Figure 3. We observe that this results in effects ranging from 0.1 to 0.5.

In addition, even small-sized effects can lead to considerable consequences of practical importance (Dodell-Feder & Tamir, 2018). If and when they do so is context-dependent on the specific real-world use case, including how widespread the effect is and how long it can last (Cooper, 2008). We would argue that for most AI-based applications in IS research, the number of affected people is potentially very large, as their use is also easily scalable. For example, McDonald's is using AI-based tools in recruiting to assist with over 60,000 applications in Austria alone (Danzer, 2020).

Conclusion

Explanations of decisions made by AI have been widely considered to increase users' trust and acceptance of AI-based systems. However, empirical research on different application contexts remains scarce, especially for sensitive areas, such as recruiting. In this study, we demonstrated that the effect of XAI on the acceptance of AI-based CA recruiting systems depends on contextual factors of the decision problem. Depending on their objectivity, XAI decisions can lead to a higher level of overall acceptance.

So far, research focused on technical solutions for XAI and high-level explainability examples often distanced from concrete contexts. Our results offer some first insights on the embeddedness of explanations in a specific application case where nuances of the explanations may lead to different outcomes. One can, however, imagine a host of studies that vary aspects of the situation in which explanations are embedded or that vary characteristics of the explanations themselves. For example, we tested the acceptance of CA decisions under the adverse conditions of a rejection. The influence of other decision outcomes would have to be answered by varying those outcomes in future studies. With the human bias to give preferential treatment to information or arguments that they agree with—the my-side bias (Nickerson, 1998)—one can assume that positive CA decisions will be evaluated more favorably.

The fact that explanations based on quantifiable data increase acceptance more than those based on data about “human factors” is interesting beyond the point of the variability itself. This result follows a pattern observed in the use of standardized tests in HR processes and is consistent with results showing that AI taking over jobs that require human elements is seen as worse compared to AI taking over cognition-focused jobs (Waytz & Norton, 2014). However, applications of AI in this intrinsically human dimension in the areas of emotions and soft skills have considerably improved (Mairesse et al., 2007; He, 2018; Zhou et al., 2019; Khosla et al., 2016). Currently, it is not clear how algorithm aversion depends on familiarization effects (Berger et al., 2021), and the open question is if, how, and when this familiarization happens and how it influences the acceptance of AI.

Notes

¹ For a literature overview on this distinction and its relation to human and algorithmic decision-making, see Waytz and Norten (2014).

² We do not provide a specific job or any concrete information about the field of application to our subjects to allow for a context-free evaluation.

³ <https://www.unipark.com/en/>, accessed on 22 December 2021.

⁴ More information on the subject pool and the offline recruitment can be found under <https://norstatgroup.com/offering/online-data-collection>, accessed on 22 December 2021. The ISO 26362 norm regulates the implementation and quality criteria of online access panels.

⁵ Models A1 to A3 adding the control variables in two blocks and their comparisons with the previous models as well as a NULL model can be found in the Appendix Table A.3.

⁶ Data are available from the authors upon request.

References

- Adadi, A., & Berrada, M. (2018). Peeking inside the black-box: A survey on explainable artificial intelligence (XAI). *IEEE Access*, 6, 52138–52160. <https://doi.org/10.1109/ACCESS.2018.2870052>
- Angerschmid, A., Zhou, J., Theuermann, K., Chen, F., & Holzinger, A. (2022). Fairness and explanation in AI-informed decision making. *Machine Learning and Knowledge Extraction*, 4, 556–579. <https://doi.org/10.3390/make4020026>
- Araujo, T., Helberger, N., Kruijemeier, S., & de Vreese, C. H. (2020). In AI we trust? Perceptions about automated decision-making by artificial intelligence. *AI & Society*, 35, 611–623. <https://doi.org/10.1007/s00146-019-00931-w>
- Armstrong, M., & Taylor, S. (2020). *Armstrong's handbook of human resource management practice* (15th ed.). KoganPage.
- Aslan, A., Greve, M., & Lembcke, T.-B. (2022). Let's do our bit: How information systems research can contribute to ethical artificial intelligence. *Proceedings of the AMCIS 2022*, Minneapolis, 1–10.
- Barredo Arrieta, A., Díaz-Rodríguez, N., Del Ser, J., Bennetot, A., Tabik, S., Barbado, A., Garcia, S., Gil-Lopez, S., Molina, D., Benjamins, R., Chatila, R., & Herrera, F. (2020). Explainable artificial intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI. *Information Fusion*, 58, 82–115. <https://doi.org/10.1016/j.inffus.2019.12.012>
- Bates, D., Mächler, M., Bolker, B., & Walker, S. (2015). Fitting linear mixed-effects models using lme4. *Journal of Statistical Software*, 67, 1–48. <https://doi.org/10.18637/jss.v067.i01>
- Berger, B., Adam, M., Rühr, A., & Benlian, A. (2021). Watch me improve—Algorithm aversion and demonstrating the ability to learn. *Business & Information Systems Engineering*, 63, 55–68. <https://doi.org/10.1007/s12599-020-00678-5>
- Bies, R. J., & Shapiro, D. L. (1988). Voice and justification: Their influence on procedural fairness judgments. *Academy of Management Journal*, 31, 676–685. <https://doi.org/10.5465/256465>
- Binns, R., van Kleek, M., Veale, M., Lyngs, U., Zhao, J., & Shadbolt, N. (2018). 'It's reducing a human being to a percentage': Perceptions of justice in algorithmic decisions. *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, Montreal, 1–14. <https://doi.org/10.1145/3173574.3173951>
- Black, J. S., & van Esch, P. (2020). AI-enabled recruiting: What is it and how should a manager use it? *Business Horizons*, 63, 215–226. <https://doi.org/10.1016/j.bushor.2019.12.001>
- Black, J. S., & van Esch, P. (2021). AI-enabled recruiting in the war for talent. *Business Horizons*, 64, 513–524. <https://doi.org/10.1016/j.bushor.2021.02.015>
- Burton, J. W., Stein, M.-K., & Jensen, T. B. (2020). A systematic review of algorithm aversion in augmented decision making. *Journal of Behavioral Decision Making*, 33, 220–239. <https://doi.org/10.1002/bdm.2155>
- Castelo, N., Bos, M. W., & Lehmann, D. R. (2019). Task-dependent algorithm aversion. *Journal of Marketing Research*, 56, 809–825. <https://doi.org/10.1177/0022243719851788>
- Castelvecchi, D. (2016). Can we open the black box of AI? *Nature*, 538, 20–23. <https://doi.org/10.1038/538020a>
- CIPD. (2020). *Labour market outlook: Autumn 2020*. https://www.cipd.org/globalassets/media/comms/news/9labour-market-outlook-autumn-2020_tcm18-86411.pdf
- Cooper, H. (2008). The search for meaningful ways to express the effects of interventions. *Child Development Perspectives*, 2, 181–186. <https://doi.org/10.1111/j.1750-8606.2008.00063.x>

- Danzer, A. (2020, October 17). *Grazer KI-Start-up Leftshift One hilft McDonald's, Bewerber fair auszuwählen*. Der Standard. Retrieved August 31, 2022, from <https://www.derstandard.de/story/2000120940707/grazer-ki-start-upleftshift-one-hilft-mcdonalds-bewerber-fair-auszuwaehlen>
- Datta, A., Sen, S., & Zick, Y. (2016). Algorithmic transparency via quantitative input influence: Theory and experiments with learning systems. *Proceedings of the 37th IEEE Symposium on Security and Privacy*, San Jose, 598–617. <https://doi.org/10.1109/SP.2016.42>
- DeLone, W. H., & McLean, E. R. (2003). The DeLone and McLean model of information systems success: A ten-year update. *Journal of Management Information Systems*, 19, 9–30. <https://doi.org/10.1080/07421222.2003.11045748>
- Dodell-Feder, D., & Tamir, D. I. (2018). Fiction reading has a small positive impact on social cognition: A meta-analysis. *Journal of Experimental Psychology: General*, 147(11), 1713–1727. <https://doi.org/10.1037/xge0000395>
- Dosilovic, F. K., Brcic, M., & Hlupic, N. (2018). Explainable artificial intelligence: A survey. *Proceedings of the 41st International Convention on Information and Communication Technology, Electronics and Microelectronics (MIPRO) 2018*, Opatija, 210–215. <https://doi.org/10.23919/MIPRO.2018.8400040>
- El Bekri, N., Kling, J., & Huber, M. F. (2020). A study on trust in black box models and post-hoc explanations. In F. Martínez Álvarez, A. Troncoso Lora, J. A. Sáez Muñoz, H. Quintián, & E. Corchado (Eds.), *Advances in intelligent systems and computing: 14th International Conference on Soft Computing Models in Industrial and Environmental Applications (SOCO 2019)* (Vol. 950, pp. 35–46). Springer International Publishing. https://doi.org/10.1007/978-3-030-20055-8_4
- European Commission. (2021). *Proposal for a regulation laying down harmonised rules on artificial intelligence (Artificial Intelligence Act)*. <https://digital-strategy.ec.europa.eu/en/library/proposal-regulation-laying-down-harmonised-rules-artificial-intelligence-artificial-intelligence>
- Förster, M., Klier, M., Kluge, K., & and Sigler, I. Evaluating explainable artificial intelligence – What users really appreciate. *Proceedings of the 2020 European Conference on Information Systems*, Marrakech. https://aisel.aisnet.org/ecis2020_rp/195
- Franke, T., Attig, C., & Wessel, D. (2019). A personal resource for technology interaction: Development and validation of the affinity for technology interaction (ATI) scale. *International Journal of Human-Computer Interaction*, 35, 456–467. <https://doi.org/10.1080/10447318.2018.1456150>
- Funder, D. C., & Ozer, D. J. (2019). Evaluating effect size in psychological research: Sense and nonsense. *Advances in Methods and Practices in Psychological Science*, 2, 156–168. <https://doi.org/10.1177/2515245919847202>
- Gaudiello, I., Zibetti, E., Lefort, S., Chetouani, M., & Ivaldi, S. (2016). Trust as indicator of robot functional and social acceptance. An experimental study on user conformation to iCub answers. *Computers in Human Behavior*, 61, 633–655. <https://doi.org/10.1016/j.chb.2016.03.057>
- Ghazizadeh, M., Peng, Y., Lee, J. D., & Boyle, L. N. (2012). Augmenting the technology acceptance model with trust: Commercial drivers' attitudes towards monitoring and feedback. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, Boston, 2286–2290. <https://doi.org/10.1177/1071181312561481>
- Gilliland, S. W. (1994). Effects of procedural and distributive justice on reactions to a selection system. *Journal of Applied Psychology*, 79, 691–701. <https://doi.org/10.1037/0021-9010.79.5.691>
- Gneezy, U., & Imas, A. (2017). Lab in the field. In *Handbook of economic field experiments* (Vol. 1, pp. 439–464). Elsevier. <https://doi.org/10.1016/bs.hefe.2016.08.003>
- Goodhue, D. L., & Thompson, R. L. (1995). Task-technology fit and individual performance. *MIS Quarterly*, 19, 213. <https://doi.org/10.2307/249689>
- Goodman, B., & Flaxman, S. (2017). European Union regulations on algorithmic decision-making and a “right to explanation.” *AI Magazine*, 38, 50–57. <https://doi.org/10.1609/aimag.v38i3.2741>
- Guidotti, R., Monreale, A., & Pedreschi, D. (2019). The AI black box explanation problem. *ERCIM NEWS*, 116, 12–13.
- Hoff, K. A., & Bashir, M. (2015). Trust in automation: Integrating empirical evidence on factors that influence trust. *Human Factors*, 57, 407–434. <https://doi.org/10.1177/0018720814547570>
- Hong, W., Chan, F. K. Y., Thong, J. Y. L., Chasalow, L. C., & Dhillon, G. (2014). A framework and guidelines for context-specific theorizing in information systems research. *Information Systems Research*, 25, 111–136. <https://doi.org/10.1287/isre.2013.0501>
- Hunkenschroer, A. L., & Luetge, C. (2022). Ethics of AI-enabled recruiting and selection: A review and research agenda. *Journal of Business Ethics*, 178, 977–1007. <https://doi.org/10.1007/s10551-022-05049-6>
- Huysmans, J., Dejaeger, K., Mues, C., Vanthienen, J., & Baesens, B. (2011). An empirical evaluation of the comprehensibility of decision table, tree and rule based predictive models. *Decision Support Systems*, 51, 141–154. <https://doi.org/10.1016/j.dss.2010.12.003>

- Jacovi, A., Marasović, A., Miller, T., & Goldberg, Y. (2021). Formalizing trust in artificial intelligence. *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, Virtual Conference, 624–635. <https://doi.org/10.1145/3442188.3445923>
- Jian, J.-Y., Bisantz, A. M., & Drury, C. G. (2000). Foundations for an empirically determined scale of trust in automated systems. *International Journal of Cognitive Ergonomics*, 4, 53–71. https://doi.org/10.1207/S15327566IJCE0401_04
- Jussupow, E., Benbasat, I., & Heinzl, A. (2020). Why are we averse towards algorithms? A comprehensive literature review on algorithm aversion. *Proceedings of the 2020 European Conference on Information Systems*, Virtual Conference. https://aisel.aisnet.org/ecis2020_rp/168
- Kießling, S., Figl, K., & Remus, U. (2021). Human experts or artificial intelligence? Algorithm aversion in fake news detection. *Proceedings of the 2021 European Conference on Information Systems*, Marrakech. https://aisel.aisnet.org/ecis2021_rp/149
- Kim, B., Park, J., & Suh, J. (2020). Transparency and accountability in AI decision support: Explaining and visualizing convolutional neural networks for text information. *Decision Support Systems*, 134, 113302. <https://doi.org/10.1016/j.dss.2020.113302>
- Kuznetsova, A., Brockhoff, P. B., & Christensen, R. H. B. (2017). lmerTest package: Tests in linear mixed effects models. *Journal of Statistical Software*, 82. <https://doi.org/10.18637/jss.v082.i13>
- Langer, M., König, C. J., & Papathanasiou, M. (2019). Highly automated job interviews: Acceptance under the influence of stakes. *International Journal of Selection and Assessment*, 27, 217–234. <https://doi.org/10.1111/ijsa.12246>
- Lee, J. D., & See, K. A. (2004). Trust in automation: Designing for appropriate reliance. *Human Factors*, 46, 50–80. https://doi.org/10.1518/hfes.46.1.50_30392
- Lee, M. K. (2018). Understanding perception of algorithmic decisions: Fairness, trust, and emotion in response to algorithmic management. *Big Data & Society*, 5(1). <https://doi.org/10.1177/2053951718756684>
- Lehmann, C., Haubitz, C., Fuegener, A., & Thonemann, U. (2020). Keep it mystic? – The effects of algorithm transparency on the use of advice. *Proceedings of the 2020 International Conference on Information Systems*, Hyderabad. https://aisel.aisnet.org/icis2020/hci_artintel/hci_artintel/3
- Lenth, R., Singmann, H., Love, J., Buerkner, P., & Herve, M. (2020). Estimated Marginal Means, aka Least-Squares Means. R package version 1.4.7 (Version 1.4.8) [Computer software]. Retrieved August 31, 2022, from <https://CRAN.R-project.org/package=emmeans>
- Letham, B., Rudin, C., McCormick, T. H., & Madigan, D. (2015). Interpretable classifiers using rules and Bayesian analysis: Building a better stroke prediction model. *Annals of Applied Statistics*, 9, 1350–1371. <https://doi.org/10.1214/15-AOAS848>
- Mädche, A., Legner, C., Benlian, A., Berger, B., Gimpel, H., Hess, T., Hinz, O., Morana, S., & Söllner, M. (2019). AI-based digital assistants. *Business & Information Systems Engineering*, 61, 535–544. <https://doi.org/10.1007/s12599-019-00600-8>
- McGill, T. J., & Klobas, J. E. (2009). A task–technology fit view of learning management system impact. *Computers & Education*, 52, 496–508. <https://doi.org/10.1016/j.compedu.2008.10.002>
- McKinsey. (2021). *The state of AI in 2021*. McKinsey Analytics. Retrieved August 31, 2022, from <https://www.mckinsey.com/business-functions/quantumblack/our-insights/global-survey-the-state-of-ai-in-2021>
- Nickerson, R. S. (1998). Confirmation bias: A ubiquitous phenomenon in many guises. *Review of General Psychology*, 2, 175–220. <https://doi.org/10.1037/1089-2680.2.2.175>
- Ochmann, J., Michels, L., Zilker, S., Tiefenbeck, V., & Laumer, S. (2020). The influence of algorithm aversion and anthropomorphic agent design on the acceptance of AI-based job recommendations. *Proceedings of the 2020 International Conference on Information Systems*, Hyderabad. https://aisel.aisnet.org/icis2020/is_workplace_fow/is_workplace_fow/4
- Ochmann, J., Zilker, S., & Laumer, S. (2020). Job seekers' artificial intelligence-related black box concerns. *Proceedings of the 2020 Computers and People Research Conference*, Nürnberg. Advance online publication. <https://doi.org/10.1145/3378539.3393841>
- Peters, D., Vold, K., Robinson, D., & Calvo, R. A. (2020). Responsible AI—Two frameworks for ethical design practice. *IEEE Transactions on Technology and Society*, 1, 34–47. <https://doi.org/10.1109/TTS.2020.2974991>
- Pöhler, G., Heine, T., & Deml, B. (2016). Itemanalyse und Faktorstruktur eines Fragebogens zur Messung von Vertrauen im Umgang mit automatischen Systemen. *Zeitschrift Für Arbeitswissenschaft*, 70, 151–160. <https://doi.org/10.1007/s41449-016-0024-9>

- Ramon, Y., Vermeire, T., Martens, D., Evgeniou, T., & Toubia, O. (2021). How should artificial intelligence explain itself? Understanding preferences for explanations generated by XAI algorithms. *SSRN Electronic Journal*. Advance online publication. <https://doi.org/10.2139/ssrn.3877426>
- Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). Why should I trust you? Explaining the predictions of any classifier. *Proceedings of the 2020 ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, San Francisco. Advance online publication. <https://doi.org/10.1145/2939672.2939778>
- Rudin, C. (2019). Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature Machine Intelligence*, 1, 206–215. <https://doi.org/10.1038/s42256-019-0048-x>
- Samek, W. (2020). Learning with explainable trees. *Nature Machine Intelligence*, 2(1), 16–17. <https://doi.org/10.1038/s42256-019-0142-0>
- Schuler, H. (2013). Personalauswahl. In R. Stock-Homburg (Ed.), *Handbuch strategisches personalmanagement* (pp. 29–58). Springer Fachmedien Wiesbaden. https://doi.org/10.1007/978-3-658-00431-6_3
- Selbst, A. D., & Powles, J. (2017). Meaningful information and the right to explanation. *International Data Privacy Law*, 7, 233–242. <https://doi.org/10.1093/idpl/ix022>
- Seong, Y., & Bissantz, A. M. (2008). The impact of cognitive feedback on judgment performance and trust with decision aids. *International Journal of Industrial Ergonomics*, 38, 608–625. <https://doi.org/10.1016/j.ergon.2008.01.007>
- Simperl, E., & Mochol, M. (2009). A case study in building semantic eRecruitment applications. In R. García (Ed.), *Semantic web for business: Cases and applications* (pp. 83–104). Information Science Reference. <https://doi.org/10.4018/978-1-60566-066-0.ch005>
- Title 1.81.8. Automated Decision Systems Accountability Act of 2020 (2020). *AB-2269 personal rights: automated decision systems*. California Legislature. Retrieved August 31, 2022, from https://leginfo.ca.gov/faces/billTextClient.xhtml?bill_id=201920200AB2269
- Toreini, E., Aitken, M., Coopamootoo, K., Elliott, K., Zelaya, C. G., & van Moorsel, A. (2020). The relationship between trust in AI and trustworthy machine learning technologies. *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*, Barcelona, 272–283. <https://doi.org/10.1145/3351095.3372834>
- Truxillo, D. M., Bodner, T. E., Bertolino, M., Bauer, T. N., & Yonce, C. A. (2009). Effects of explanations on applicant reactions: A meta-analytic review. *International Journal of Selection and Assessment*, 17, 346–361. <https://doi.org/10.1111/j.1468-2389.2009.00478.x>
- Venkatesh, V., Morris, M. G., Davis, G. B., & Davis, F. D. (2003). User acceptance of information technology: Toward a unified view. *MIS Quarterly*, 27, 425–478.
- Vreede, T. de, Raghavan, M., & Vreede, G.-J. de (2021). Design foundations for AI assisted decision making: A self determination theory approach. *Journal of Artificial Intelligence Research*. Advance online publication. <https://doi.org/10.24251/HICSS.2021.019>
- Waytz, A., & Norton, M. I. (2014). Botsourcing and outsourcing: Robot, British, Chinese, and German workers are for thinking—not feeling—jobs. *Emotion*, 14, 434–444. <https://doi.org/10.1037/a0036054>
- Yang, X., & Aurisicchio, M. (2021). Designing conversational agents: A self-determination theory approach. *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, Yokohama, 1–16. <https://doi.org/10.1145/3411764.3445445>
- Yeomans, M., Shah, A., Mullainathan, S., & Kleinberg, J. (2019). Making sense of recommendations. *Journal of Behavioral Decision Making*, 32, 403–414. <https://doi.org/10.1002/bdm.2118>

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Appendix

Additional Tables and Figures

Table A.1. Scale and Item Overview

Scale (Cronbach's Alpha)	Item	Mean	SD
ATI (0.86)	I like to occupy myself in greater detail with technical systems.	3.09	1.23
	I like testing the functions of new technical systems.	2.80	1.26
	I predominantly deal with technical systems because I have to.*	2.58	1.20
	When I have a new technical system in front of me, I try it out intensively.	2.89	1.24
	I enjoy spending time becoming acquainted with a new technical system.	3.07	1.26
	It is enough for me that a technical system works; I don't care how or why.*	3.13	1.21
	I try to understand how a technical system exactly works.	2.85	1.19
	It is enough for me to know the basic functions of a technical system.*	3.19	1.12
TRUST_AI (0.87)	I try to make full use of the capabilities of a technical system.	2.92	1.20
	AI systems are deceptive.*	2.80	0.89
	AI systems behave in an underhanded manner.*	3.14	0.93
	I am suspicious of the AI system's decision.*	3.05	0.98
	I am wary of AI systems.*	3.58	1.01
	The use of an AI system has negative effects.*	2.83	0.92
	I am confident in AI systems.	3.18	0.97
	AI systems provide security	3.01	0.92
	AI systems have integrity.	2.85	0.91
	AI systems are dependable.	2.92	0.92
	AI systems are reliable.	3.04	0.91
	I can trust AI systems.	3.12	0.93
	I am familiar with AI systems.	3.55	1.04

* Negatively worded items, marked with *, have been reverse coded.

Robustness Check: Order Effects

Although order effects were mitigated by presenting the vignettes in random order, we cannot exclude the possibility of a learning effect over time. To investigate this possibility, we reproduce Model A1 from Table A3, as it contains only the scenario dummies, using only data from the first vignette that participants saw. This resembles a between-subject design in which each participant only sees one scenario. Thus, OLS models can be estimated as no nesting of decisions within participants must be considered.

Table A2 shows the model R1 based on those 490 observations, and the coefficients are qualitatively similar to Model A1 in Table A3, although the coefficients are not significantly different from zero due to reduced power.

When we compare Model A1 to a NULL model containing only the subject random effects using the AIC criteria, as we use a maximum likelihood estimator,

we find that AIC is reduced by 159.99 to 8295.01 in Model A1 from 8455.00 in the NULL model. In accordance, also a likelihood ratio test yields a significant result ($p < 0.001$).

We find that the inclusion of the additional predictors in Model A2 decreases AIC to 8142.861 from 8295.013 in Model 1, and the likelihood ratio test comparing Model A1 and Model A2 is also significant ($p < 0.001$).

Our final Model A3 adds the AIC for Model 3 increases to 8146.189 compared to Model 2 (AIC can increase as it penalizes model complexity). Comparing Model A2 with Model A3 using a likelihood ratio test yields no significant result. Summing up, these results indicate that the included sociodemographic characteristics together do not influence the overall acceptance of AI-based CA recruiting systems in recruiting, except for a small negative effect of age.

Table A.2. OLS Regression Model Based on the First Decision Only

EXPLAIN Scenarios (n)		Model R1
VERIFIABLE	LIST (77)	0.346 (0.168)
	COMPARE (59)	0.267 (0.254)
	TREE (79)	0.122 (0.235)
SOFT	LIST (70)	0.136 (0.243)
	COMPARE (67)	0.225 (0.246)
	TREE (62)	0.084 (0.251)
Intercept		3.587 (0.168)***
AIC		381.357
Number of Observations		490

***p < 0.001. The table displays the unstandardized coefficients of a multilevel regression model with subject random effects estimated with the Maximum Likelihood estimator and the overall acceptance scale as dependent variable. Standard errors are displayed in parentheses. Dummy variables are included for all six EXPLAIN scenarios with BASE serving as reference category.

Table A.3. Multilevel Regression Models on Overall Acceptance of CAs Preselection Decision

EXPLAIN Vignette		Model A1	Model A2	Model A3
VERIFIABLE	LIST	0.338 (8.431)***	0.338 (8.431)***	0.338 (8.431)***
	COMPARE	0.458 (11.439)***	0.458 (11.439)***	0.458 (11.439)***
	INTERPRET	0.382 (9.532)***	0.382 (9.532)***	0.382 (9.532)***
SOFT	LIST	0.161 (4.027)***	0.161 (4.027)***	0.161 (4.027)***
	COMPARE	0.231 (5.781)***	0.231 (5.781)***	0.231 (5.781)***
	INTERPRET	0.195 (4.873)***	0.195 (4.873)***	0.195 (4.873)***
Controls				
Affinity for Technology			-0.020 (-0.282)	-0.021 (-0.282)
Trust in AI			1.220 (12.286)***	0.669 (12.077)***
Gender (1 = Female)				-0.005 (-0.040)
Country (1 = Austria)				-0.019 (-0.165)
Age				-0.007 (-2.089)*
Income				-0.007 (-0.307)
(Intercept)		3.490 (50.621)***	-1.000 (-3.542)***	-0.547 (-1.284)
AIC		8295.013	8142.861	8146.189
ΔAIC		-159.993	-152.152	3.328
Log Likelihood		-4138.507	-4060.431	-4058.059
Number of Decisions		3430	3430	3430
Number of Participants		490	490	490

***p < 0.001; **p < 0.01; *p < 0.05. The table displays the unstandardized coefficients of multilevel regression models with subject random effects estimated with the Maximum Likelihood estimator and the overall acceptance scale as dependent variable. t-values are displayed in parentheses. Dummy variables are included for all six EXPLAIN scenarios with BASE serving as reference category; the dummy variables take the value of one for subjects' answers in the vignette referenced in the variable

name and zero for all other observations. The regression coefficient of a scenario dummy can be interpreted as the average difference in overall acceptance of this vignette to the reference vignette base. AIC = Akaike information criterion. For Model A1 Δ AIC displays the change of AIC from the NULL model containing only the random effects to Model A1. For Model A2 (A3) Δ AIC contains the difference of this model's AIC to Model A1 (A2). AIC of the NULL model is 8455.006 and Log Likelihood = -4224.5.

Robustness Check: Power Analysis Simulation

While the regression coefficients from Model R1 are qualitatively similar to those of Model A1, they are insignificant. We attribute this to reduced statistical power and thus run simulations for a between-subject design. Concretely, we use the mean and standard deviations from Table 2 in the manuscript as parameters of a normal distribution and draw overall acceptance values for sample sizes ranging from 210 to 2,310 in steps of 210. We simulate 1,500 drawings per sample size and report the share of samples where t-tests of an OLS regression for a scenario dummy yield a p-value below .05 (i.e., the realized power for an alpha level of .05 for this scenario). The

simulation results, displayed in Figure A.1, again show the pattern that the regression coefficients of the VERIFIABLE scenarios are more likely to be identified as significantly different from zero than those of the SOFT scenarios. For the strongest effect of VERIFIABLE_COMPARE we find that a sample size of around 1,200 (i.e., around 170 independent observations per scenario) is sufficient to achieve a power of .8. For the SOFT scenarios, even sample sizes of above 2,000 are not enough. This supports our conclusion that the VERIFIABLE scenarios have a stronger effect than the SOFT scenarios and also provides evidence against the results from the within-subject experiment being erroneous due to overpowering.

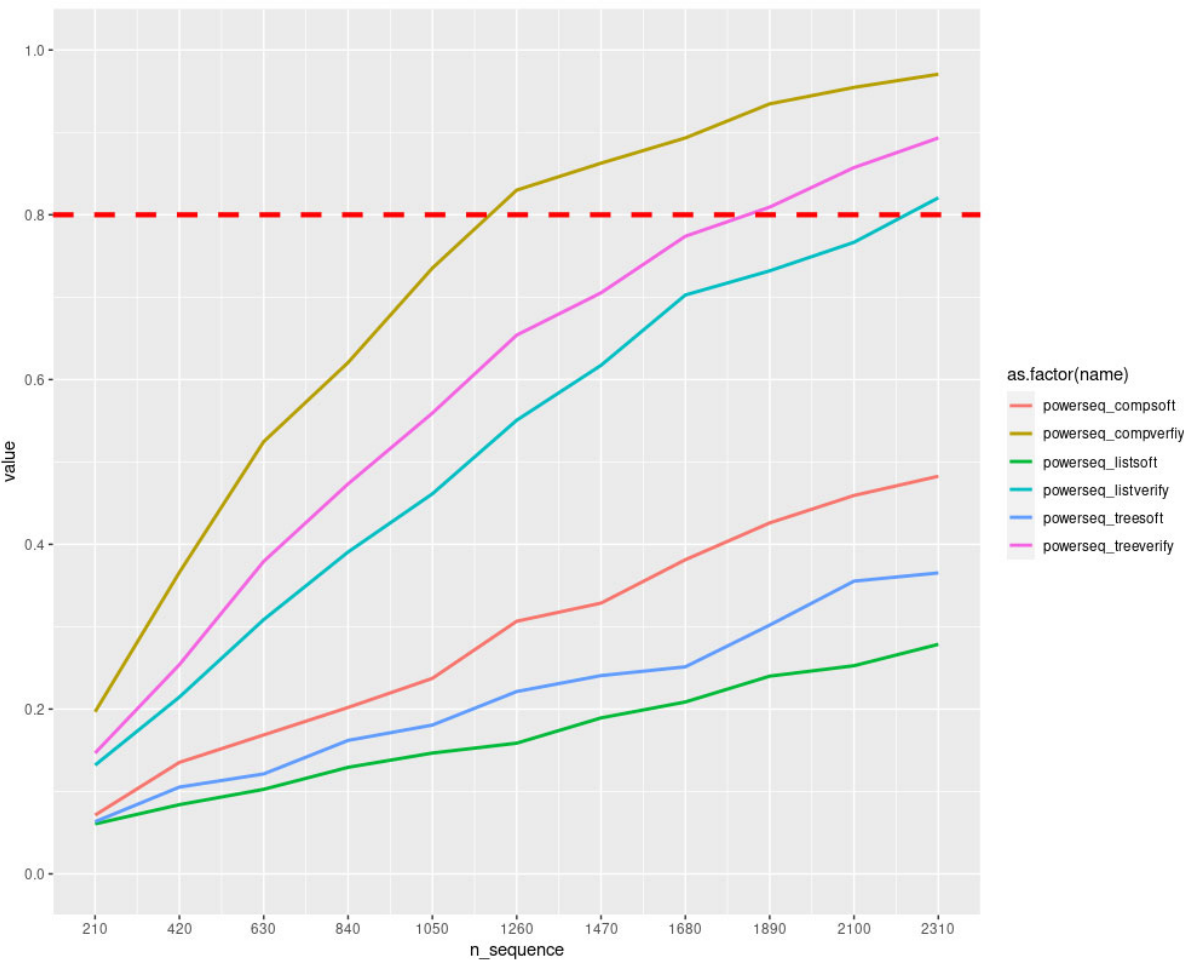


Figure A.1. Results of Power Simulation

Robustness Checks: Exclusion of Slowest and Fastest 5 Percent

Table A.4. Descriptive Statistics for Dependent Variables by Scenario Including Fast and Slow Completing Subjects

Dependent Variables	BASE No Info	EXPLAIN					
		LIST		COMPARE		INTERPRET	
		VERIFIABLE	SOFT	VERIFIABLE	SOFT	VERIFIABLE	SOFT
Applyplus	3.48 (1.72)	3.98 (1.69)	3.82 (1.70)	4.08 (1.72)	3.86 (1.75)	4.03 (1.64)	3.82 (1.70)
Choosebot	3.44 (1.76)	3.67 (1.76)	3.56 (1.78)	3.79 (1.78)	3.59 (1.76)	3.70 (1.71)	3.60 (1.76)
Goodbot	3.60 (1.69)	3.87 (1.70)	3.73 (1.70)	4.04 (1.74)	3.81 (1.72)	3.92 (1.70)	3.77 (1.70)
Trust	3.46 (1.56)	3.84 (1.62)	3.58 (1.62)	3.94 (1.62)	3.73 (1.64)	3.89 (1.61)	3.68 (1.63)
Fair	3.65 (1.48)	4.00 (1.52)	3.77 (1.49)	4.09 (1.53)	3.88 (1.55)	4.03 (1.52)	3.82 (1.53)
Overall	3.53(1.48)	3.87 (1.53)	3.69 (1.52)	3.99 (1.54)	3.77 (1.57)	3.91 (1.50)	3.74 (1.54)
Alpha	0.94	0.95	0.95	0.95	0.96	0.95	0.95

Notes: The table displays the mean values (standard deviations in parentheses) for all dependent variables in each scenario. All items range from one to seven, with higher values indicating more trust, agreement etc. "Overall" refers to the overall acceptance scale and is computed as the mean value of the behavioral intention, trust, and fairness measures. The bottom row labelled "Alpha" contains the Cronbach's alpha values for the overall acceptance scale in each of the scenarios separately, with all values indicating very good consistency. All values are based on n=554 participants.

Table A.5. Multilevel Model Including Fast and Slow Completing Subjects

EXPLAIN Scenarios		Model R2
VERIFIABLE	LIST	0.349 (9.188)***
	COMPARE	0.463 (12.203)***
	TREE	0.387 (10.206)***
SOFT	LIST	0.166 (4.366)***
	COMPARE	0.249 (6.553)***
	TREE	0.213 (5.621)***
	(Intercept)	3.526 (54.416)***
AIC		9427.119
ΔAIC		-182.347
Log Likelihood		-4704.559
Number of Decisions		3878
Number of Participants		554

***p < 0.001. The table displays multilevel regression models with subject random effects estimated with the Maximum Likelihood estimator and the overall acceptance scale as dependent variable. t-values are displayed in parentheses. Dummy variables are included for all six EXPLAIN scenarios with BASE serving as reference category. ΔAIC displays the change of AIC from the NULL model containing only the random effects to Model 1. The AIC of the NULL model is 9609.465 and Log Likelihood = -4801.7.

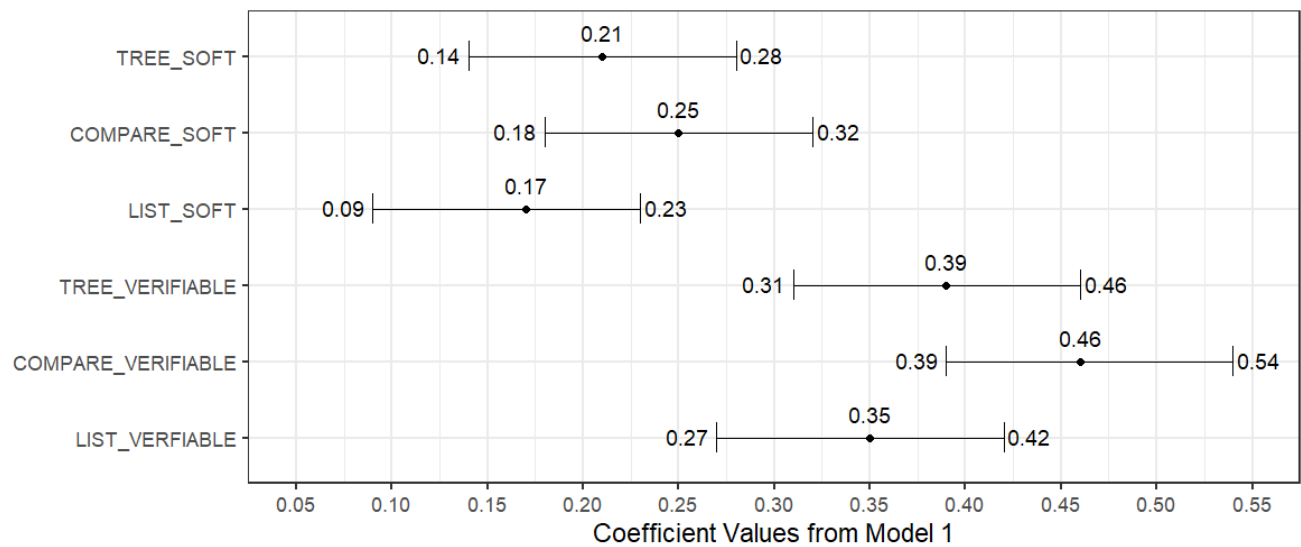


Figure A.2. Coefficients and Confidence Intervals for Scenario Dummies Including Fast and Slow Completing Subjects

Note: The figure displays the regression coefficients of the scenario dummies from Model R2 with the corresponding 95-percent confidence intervals, which were computed using 1,000 parametric bootstrap samples.

Table A.6. Multilevel Model Including Fast and Slow Completing Subjects

				EXPLAIN					
				LIST		COMPARE		INTERPRET	
				VERIFIABLE	SOFT	VERIFIABLE	SOFT	VERIFIABLE	SOFT
EXPLAIN	LIST	VERIFIABLE	< 0.001	x					
		SOFT	< 0.001	< 0.001	x				
	COMPARE	VERIFIABLE	< 0.001	0.042	< 0.001	X			
		SOFT	< 0.001	0.117	0.304	< 0.001	x		
	INTERPRET	VERIFIABLE	< 0.001	0.950	< 0.001	0.418	0.050	x	
		SOFT	< 0.001	0.007	0.873	< 0.001	0.968	< 0.001	x

Note: The table presents the Tukey-adjusted p-values of pairwise comparisons between all conditions computed using the R package emmeans (Lenth et al., 2019).

English Translation of the Original German Questionnaire

Text **highlighted in yellow** indicates additional notes that are not included in the questionnaire.

In which region of the country do you live?

Wien

Niederösterreich

Burgenland

Oberösterreich

Salzburg

Steiermark

Kärnten

Tirol

Vorarlberg

Sex

Male

Female

What level of education currently applies to you?

Mandatory Education

Apprenticeship

VET school

High School

University

How old are you?

Which employment situation mainly applies to you?

Full Time Job

Part Time Job

Partial Retirement

Minor Employment

Unemployed

Military Service

Voluntary Social Year

Parental Leave

Without Occupation

In School / Training

At the beginning we would like to ask you a few questions about IT systems. IT systems are all types of electronic systems that process data. This includes computers, mobile phones and various other communication systems.

Please indicate if you agree or disagree with the following statements.

Fully agree

Agree

Neither/nor

disagree

Fully disagree

I like to occupy myself in greater detail with technical systems.

I like testing the functions of new technical systems.

I predominantly deal with technical systems because I have to.

When I have a new technical system in front of me, I try it out intensively.

I enjoy spending time becoming acquainted with a new technical system.

It is enough for me that a technical system works; I don't care how or why

I try to understand how a technical system exactly works.

It is enough for me to know the basic functions of a technical system.

I try to make full use of the capabilities of a technical system.

**Artificial intelligence is a special software solution in the field of IT systems.
Have you heard of artificial intelligence before?**

Yes

No

How would you rate your knowledge about artificial intelligence?¹

Very Good

Good

Bad

Very Bad

Artificial intelligence (AI) means computer systems that solve tasks in a way that is not previously precisely specified by humans. AI systems find solutions independently and can therefore make decisions without human involvement. AI systems are already being used today for various purposes.

Examples:

- Identification of people and objects in photos
- cancer detection
- opponents in computer games
- Recommendation of products or videos
- Propose search results
- Prediction of machine failures

Please indicate if you agree or disagree with the following statements.

Fully agree

Agree

Neither/nor

disagree

Fully disagree

AI systems are deceptive.

AI systems behave in an underhanded manner.

I am suspicious of the AI system's decision.

I am wary of AI systems.

The use of an AI system has negative effects.

I am confident in AI systems.

AI systems provide security

AI systems have integrity.

AI systems are dependable.

AI systems are reliable.

I can trust AI systems.

I am familiar with AI systems.

The next part of the questionnaire is about chatbots.

A chatbot, also called a digital assistant or virtual robot, communicates with you instead of a human, often in written form through a chat window or verbally through voice recognition. Many people interact with chatbots every day, the best-known examples are Amazon's Alexa or Apple's Siri.

¹ A programming error made this question optional, resulting in missing values.

Have you heard of chatbots before?

Yes

No

Have you ever used a chatbot?

Yes

No

In the following part of the survey, we are now interested in the use of chatbots in the recruiting process. Since chatbots work with different functions of artificial intelligence, such as pattern recognition, text recognition, image recognition, speech and emotion recognition, companies apply chatbots at both, the communication with potential applicants as well as in selecting suitable candidates. All of the following questions relate to the use of chatbots in the application process.

Please indicate if you agree or disagree with the following statements.

Fully agree

Agree

Neither/nor

disagree

Fully disagree

By using a chatbot, I expect feedback to be faster than in a common application process with a human expert.

By using a chatbot in an application process, I expect a more transparent decision than from a human expert.

By using a chatbot in an application process, I expect a fairer rating than from a human expert.

Using a chatbot in the application process increases my chances of a fair assessment.

Interacting with a chatbot is clear and understandable.

I would find a chatbot to be flexible to interact with.

Interacting with a chatbot is complicated and difficult to understand.

Interacting with a chatbot is easy for me.

If I told my friends that I was using a chatbot, it would make a good impression.

People who are important to me would encourage me to use a chatbot.

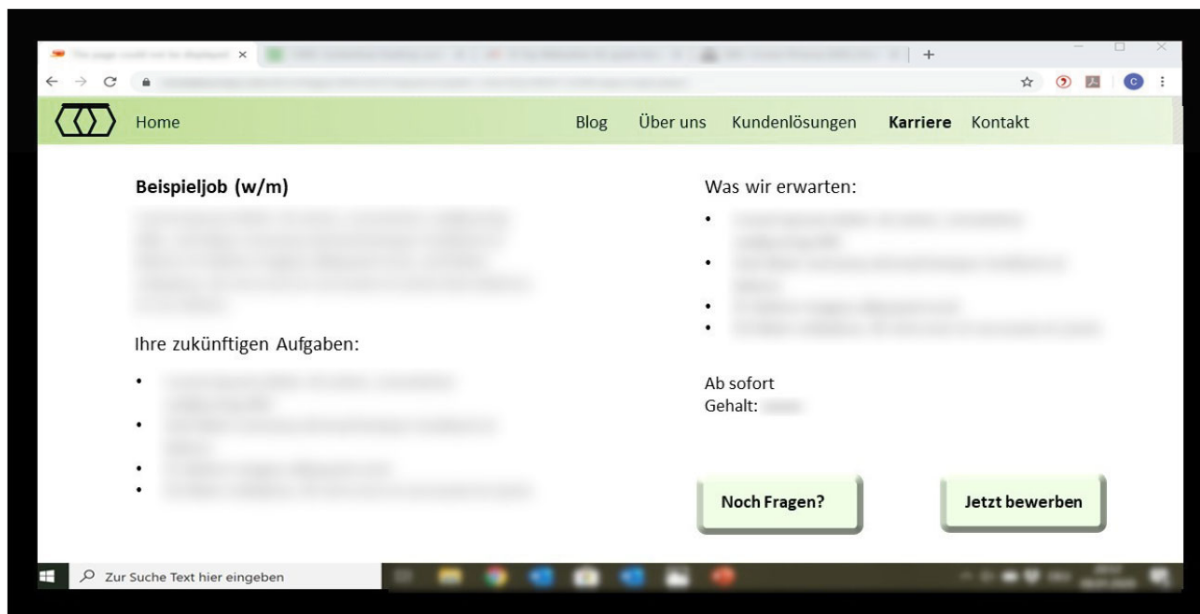
It is important to me that I can contact a person if I need help while interacting with a chatbot.

It is important to me that there are instructions on how to use the chatbot.

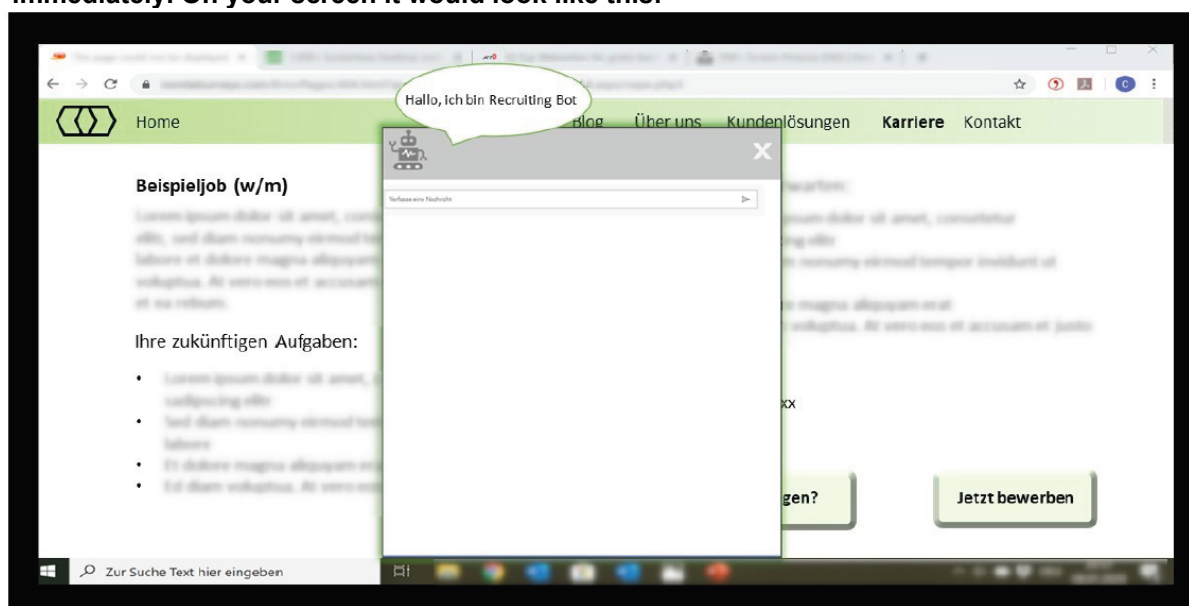
With a chatbot it is possible to communicate flexibly at every time. By using natural language, the chatbot communicates constantly and promptly with (potential) candidates by answering questions about vacancies, providing feedback and updates on the process, and communication gaps between applicants and recruiters decrease.

Please put yourself in the following situation:

You would like to apply to a company for a suitable vacancy. You will see the following job advertisement on the homepage:



You already have some questions in advance and therefore you click on the "Any questions?" Button. A chatbot called "Recruiting Bot" will appear. The chatbot will communicate with you via chat and answers your questions immediately. On your screen it would look like this:



For the specific use of a chatbot immediately answering questions about the vacancy, please indicate to what extent you agree or disagree with the following statements?.

Please indicate if you agree or disagree with the following statements.

- Fully agree*
- Mostly agree*
- Agree*
- Neither/nor*
- disagree*

Mostly disagree
Fully disagree

The use of a chatbot that immediately answers my question during the application process positively influences my decision to apply.

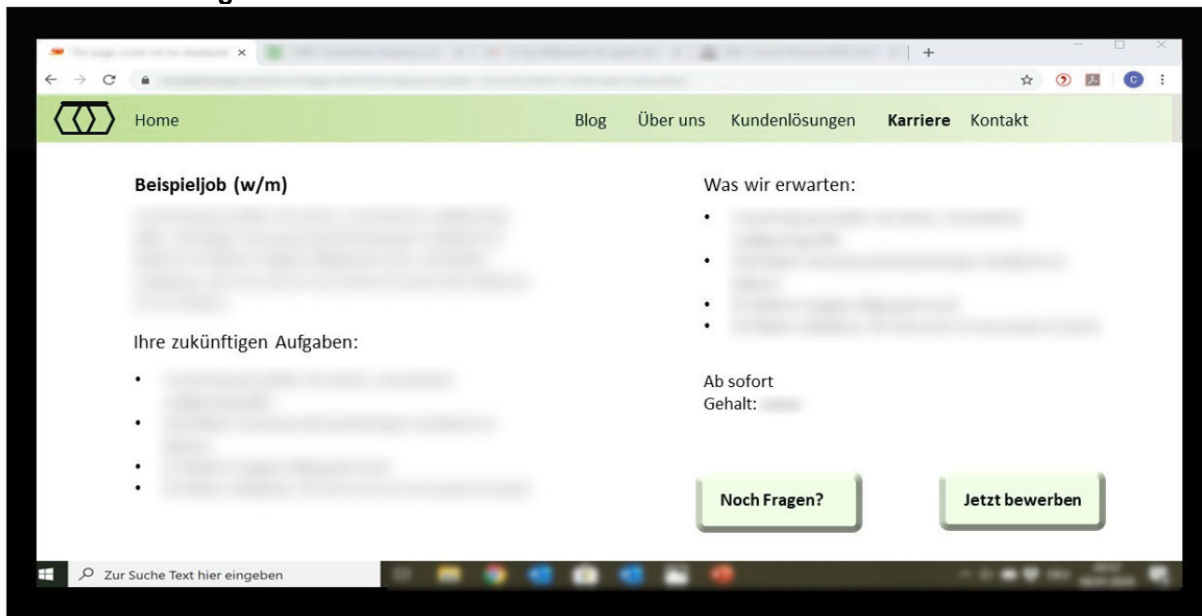
If this chatbot is available as an alternative to a human contact person, then I choose the chatbot.

I consider the use of chatbots in the application process to be positive.

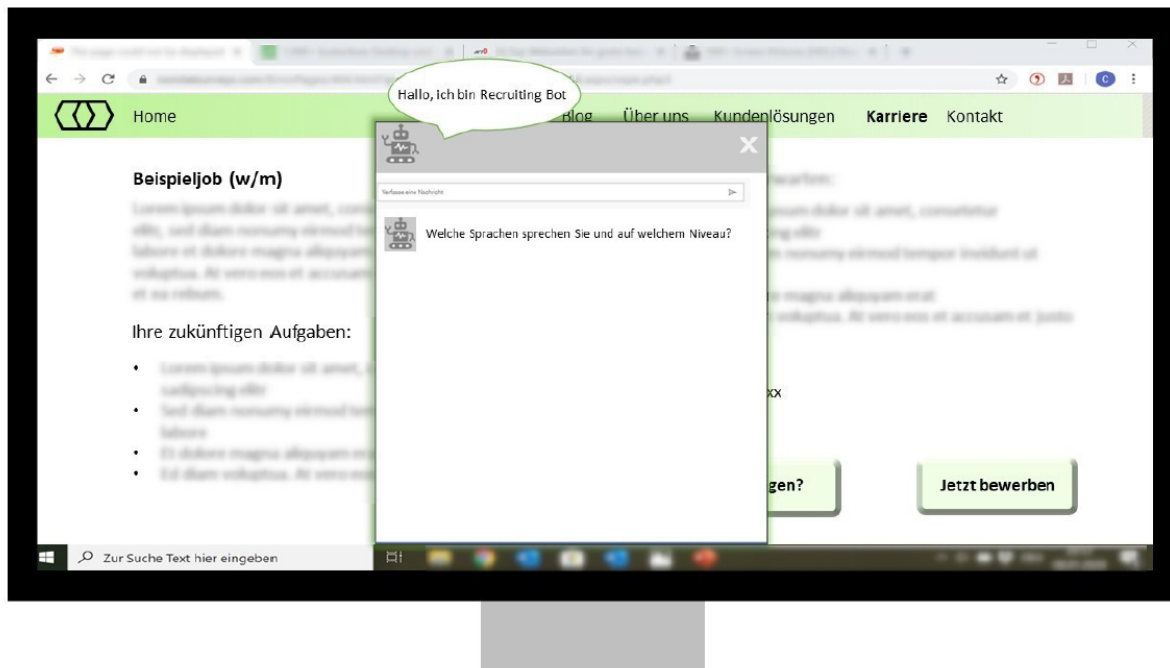
I trust that the chatbot just described will provide good answers to my questions about the vacancy.

Fully trust
Mostly trust
trust
Neither/nor
do not trust
Mostly do not trust
Fully do not trust

There are now descriptions for other situations of how chatbots can be used in the recruiting process. From now on, always imagine that you are applying for a suitable vacancy. On the homepage you will see the following job advertisement again:



The concrete situation is: You want to apply for the vacancy and therefore you click on the "Apply now" button. A chatbot called "Recruiting Bot" will appear. Instead of a human expert, the chatbot preselects the candidates. The chatbot communicates with you via chat and provides you with all the necessary questions for the recruiting process to evaluate your qualifications. On your screen it would look like this:



After this introduction, 7 scenarios were presented in random order. For all scenarios, the following questions were asked. After the questions, the scenarios are presented.

For this specific use of a chatbot in the preselection, please indicate whether you agree or disagree with the following statements.

- Fully agree
- Mostly agree
- Agree
- Neither/nor
- disagree
- Mostly disagree
- Fully disagree

The use of a chatbot that during the application process [, that explains its decision using varying according to scenario, none in BASE] positively influences my decision to apply.

If this chatbot is available as an alternative to a human interviewer, then I choose the chatbot.

I consider the use of chatbots in the application process to be positive.

It is important to me that a human checks the decision of the chatbot.

Do you trust the chatbot decision just described?

- Fully trust
- Mostly trust
- trust
- Neither/nor
- do not trust
- Mostly do not trust
- Fully do not trust

How fair or unfair is this preselection decision of the chatbot?

Fully fair
Mostly fair
Fair
Neither/nor
Unfair
Mostly unfair
Fully unfair

Scenario descriptions:

Scenario BASE:

Please imagine the following situation:

The recruiting process with the Recruiting Bot is now completed and you will immediately receive the following message:

"You have been rejected because your profile does not meet the requirements of the vacancy."

Scenario LIST_VERIFYABLE:

The application process with the Recruiting Bot is now completed and you will immediately receive the following message:

"You have been rejected because your profile does not meet the requirements of the vacancy.

After the evaluation of all criteria carried out by the recruiting bot, the following verifiable qualifications were decisive for the decision:

Critical verifiable qualifications:

- Work experience
- English proficiency
- Computer knowledge"

Scenario LIST_SOFT:

The application process with the Recruiting Bot is now completed and you will immediately receive the following message:

"You have been rejected because your profile does not meet the requirements of the vacancy.

After the evaluation of all criteria by the Recruiting Bot, the following soft skills were decisive for the decision:

Crucial soft skills:

- Teamwork
- Communication skills
- Diligence"

Scenario COMPARE_VERIFYABLE:

The application process with the Recruiting Bot is now completed and you will immediately receive the following message:

"You have been rejected because your profile does not meet the requirements of the vacancy.

After the evaluation of all criteria carried out by the Recruiting Bot, the following verifiable qualifications were decisive for the decision.

You can see from the illustration that you did significantly worse than other applicants."

Verifiable Qualifications

Work Experience	0 Years	●		●	20 Years
Command of English	None		●		Expert
Computer Knowledge	None	●		●	Expert

● You

● Average of other applicants

Scenario COMPARE_SOFT:

The application process with the Recruiting Bot is now completed and you will immediately receive the following message:

“You have been rejected because your profile does not meet the requirements of the vacancy.
After the evaluation of all criteria carried out by the Recruiting Bot, the soft skills presented below were decisive for the decision.

You can see from the illustration that you do significantly worse than other applicants.”

Soft skills	bad	average	good
Ability to work in teams			
Communication Skills			
Diligence			

 You

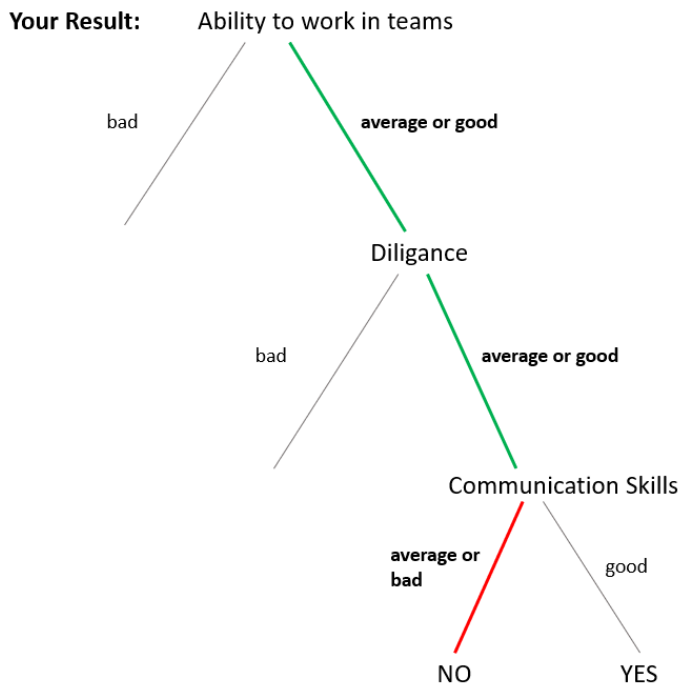
 Average of other Applicants

Scenario INTERPRET_SOFT:

The application process with the Recruiting Bot is now completed and you will immediately receive the following message:

“You have been rejected because your profile does not meet the requirements of the vacancy.
As an explanation, you will receive the chatbot’s decision-making rules, which were used for decisive soft skills.

You can see from the illustration that you fulfill the soft skills “teamwork” and “diligence” (green), but because of insufficient “communication skills” you are rejected from the recruiting process (red).



Scenario INTERPRET_HARD:

The application process with the Recruiting Bot is now completed and you will immediately receive the following message:

“You have been rejected because your profile does not meet the requirements of the vacancy.
As an explanation you will receive the decision rules of the chatbot which were used for decisive verifiable qualifications.

You can see from the illustration that you meet the verifiable qualifications “work experience” and “knowledge of English” (green), but due to insufficient “Computer skills” you are rejected from the application process (red).

